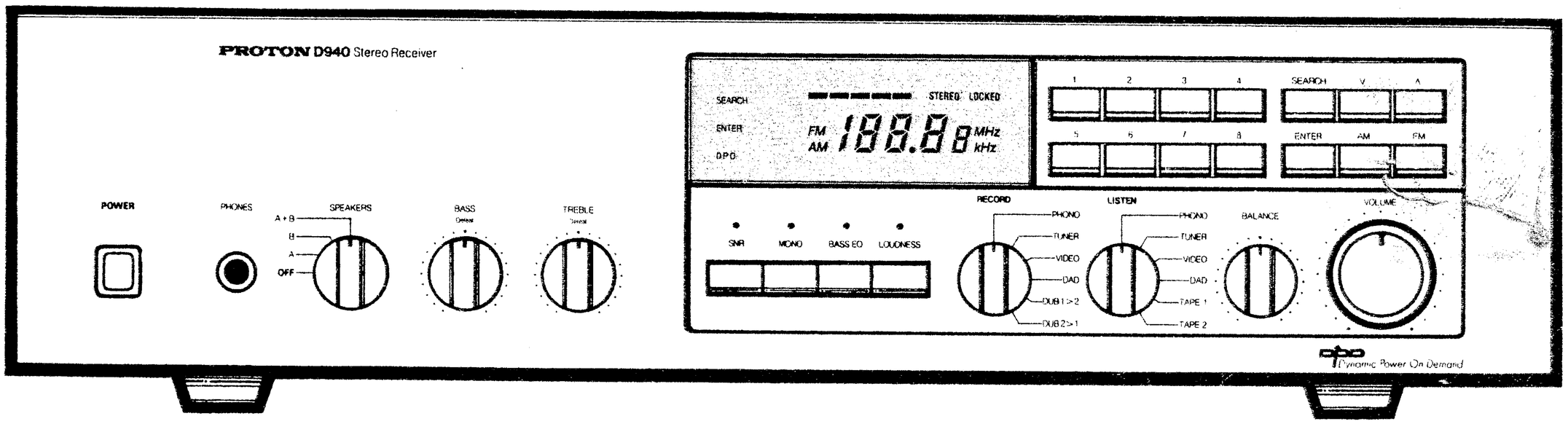
PROTON

D940 SERVICE MANUAL

STEREO RECEIVER



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SPECIFICATIONS

FM TUNER SECTION:		LIMIT	NORMAL
1. Usable sensitivity (75 OHM):		1.5uV	1.2uV
2. 50dB quieting sensitivity:	Mono/Stereo:	4.0uV/35uV	3.2uV/20uV
3. THD @ 65dBf:	Mono/Stereo 100Hz:	0.3%/0.5%	0.2%/0.3%
	Mono/Stereo 1 KHz:	0.3%/0.3%	0.1%/0.12%
	Mono/Stereo 10KHz:	0.3%/0.5%	0.2%/0.3%
4. S/N ratio (A weighted, at 65dBf)	Mono/Stereo:	78dB/68dB	83dB/72dB
5. HUM and noise @ 65dBf:	Mono/Stereo:	70dB/65dB	75dB/68dB
6. De-emphasis accuracy:	50us:	+-1dB	+-0.5dB
	75us:	+-1dB	+-0.5dB
7. Capture ratio @ 45dBf:		2.5dB	1.5dB
8. Alternated channel selectivity @ 45dBf:		65dB	71dB
9. Spurious response 1/2 IF:		75dB	82dB
10.Image rejection:		65dB	70dB
11.If rejection:		95dB	110dB
12.AM suppression @ 65dBf:		53dB	65dB
13.Stereo seperation:	100Hz:	30dB	42dB
	1 KHz:	40dB	48dB
	10 KHz:	30dB	42dB
14.Output level:		450-850mV	

AM TUNER SECTION:		
1. Usable sensitivity:	800uV/m	500uV/m
2. Selectivity:	30dB	35dB
3. Image rejection:	30dB	35dB
4. IF rejection:	45dB	52dB
5. AGC:	45dB	50dB
6. T.H.D. 400Hz 30% modulation @ 10mV:	1.0%	0.5%
7. Audio bandwidth:	200Hz--3KHz	150--3K8 Hz
8. S/N @ 10mV:	35dB	40dB
9. 2nd & 3rd tweet:	10%	3%

AMPLIFIER SECTION:

Rated Power Output (20Hz-20kHz, 8 OHM):	40W, Per Channel
Dynamic Power at 8 OHM/4 OHM/2 OHM	
1) I.H.F. Standard 20ms Duration:	180W/350W/440W
2) 100ms Duration:	170W/240W/250W
3) 200ms Duration:	160W/210W/220W
(Duty Cycle: 500ms)	
Dynamic Headroom at 8 OHM/4 OHM/2 OHM:	6dB
T.H.D. at Rated Power:	0.02%
I.M.D. at Rated Power:	0.008%
Clipping Power at 8 OHM/4 OHM/2 OHM:	50W/80W/100W
Damping Factor:	>90
Frequency Response 20-20kHz:	±0.2dB
Power Bandwidth at T.H.D. 0.1%:	10-60kHz
High Level Input Resistance/Capacitance:	40K OHM/220pF
Line Input Sensitivity (Video):	150mV
(Tape Play):	150mV
(Phono MM):	2.5mV
(Phono MC High/Low):	0.2mV/0.1mV
Residual Noise (Flat):	0.2mV
Channel Crosstalk (1 kHz):	75dB
Function Crosstalk (1 kHz):	78dB
S/N Ratio (A-Weighted) (Video):	105dB
(Tape Play):	105dB
(Phono MM):	92dB
(Phono MC):	75dB
Bass Control @ 100Hz (Boost/cut):	+9/-9dB
Treble Control @ 10kHz (Boost/cut):	+9/-9dB
Bass EQ @ 75Hz:	+3dB
Loudness @ -30dB (100Hz):	+6dB
(10kHz):	+3dB
Phono Input Resistance (MM/MC):	47K OHM/100 OHM
Phono Input Capacitance (Selectable):	100pF/200pF/320pF
Phono Overload @ 1 kHz T.H.D. 0.1% (MM):	250mV
Phono T.H.D. at 4.5V Output (MM):	0.01%
(MC):	0.03%
RIAA Response Accuracy MM/MC:	±0.2dB/±0.5dB
Dimensions W x H x D (inch):	16-1/2"x3-7/16"x12-5/8"
(cm):	42.0x8.7x32.0 CM
Net Weight:	8.7Kg/19.14Lbs

ALIGNMENT

1. FM ALIGNMENT: a. SELECTOR SWITCH AT FM POSITION
b. AC LINE VOLTAGE AT RATED OVLTAGE
c. MONITOR OUTPUT AT REC OUT

SECTION		FM	SG		DIAL SETTING	INDICATOR	ADJUSTMENT	ADJUST FOR
		CONNECTION	CARRIER	MOD				
THD	1		98MHz	100% MOD 1KHz	98MHz	DIGITAL VOLTAGE METER	IFT 101 A UP SIDE CORE	PIN TP1 VDTAGE 0+-0.05V
	2					THD METER	IFT101 B DOWN SIDE CORE	MINIMUM THD
RF	3	CONNECT TO FM 300 OHM ANTENNA TERMINAL ON THE REAL PANEL THROUGH FM ANTENNA	REPEAT STEP 1 AND 2			DIGITAL VOLTAGE METER	L005	PIN TP2 TUNING VOLTAGE 4.3+-0.1V
	1		90MHz	100% MOD 400Hz	90KHz			
	2		106MHz		106MHz		TC004	PIN TP2 TUNING VOLTAGE 18.7+-0.2V
	3		REPEAT STEP 1 AND 2					
SENS	1		90MHz	100% MOD 400Hz	90MHz	VTVM & OSCILLOSCOPE	L001-L003	MAXMIMUM OUTPUT
	2		106MHz		106MHz		TC001-TC003	
	3		98MHz		98MHz		IFT001	
	4		REPEAT STEP 1. 2 AND 3					
SIGNAL STRENGTH INDICATOR			98MHz ANT. INPUT LEVEL 70dB	100% MOD 400Hz	98MHz	VTVM & OSCILLOSCOPE	VR102	THE 5th LAMP JUST LIGHT
AUTO STOP LEVEL			98MHz ANT. INPUT LEVEL 14dB	100% MOD 400Hz	98MHz	VTVM & OSCILLOSCOPE	VR101	LOCK DISPLAY JUST LIGHT

2. FM MPX ALIGNMENT: a. SELECTOR SWITCH AT FM POSITION
b. AC LINE OVLTAGE AT RATED VOLTAGE
c. MONITOR OUTPUT AT REC OUT SG AND CONNECTED TO FM 300 OHM ANTENNA TERMINAL ON THE REAR PANEL THROUGH FM DUMMY ANTENNA

SECTION	STEP	FM SG	STEREO SG	DIAL SETTING	INDICATOR	ADJUSTMENT	ADJUST FOR
SEPERA-TION	1	98MHz	10% 19KHz (PILOT) L ONLY	98MHz	VTVM & OSCILLOSCOPE TO R REC OUT	VR103	MINIMUM OUTPUT
	2		10% 19KHz (PILOT) R ONLY	98MHz	VTVM & OSCILLOSCOPE TO L REC OUT	VR103	
	3		REPEAT STEP 1 AND 2				
	4		IF THERE IS EXCESSIVE DIFFERENCE BETWEEN LEAK-FREE EFFECT OF BOTH CHANNELS, SLIGHTLY ADJUST VR103 SO THAT THE LEVEL OF SIGNAL LEAKAGE OF BOTH CHANNELS ARE EQUAL.			
LSNR	1	98MHz ANT. INPUT LEVEL 30dB	10% 19KHz (PILOT) 1KHz 90% L+R	98MHz	VTVM & OSCILLOSCOPE	VR301	JUST MAXIMUM SEPARATION
	2	98MHz ANT. INPUT LEVEL 50dB	10% 19KHz (PILOT) 1KHz 70% L+R	98MHz	VTVM & OSCILLOSCOPE	VR302	JUST MAXIMUM SEPARATION

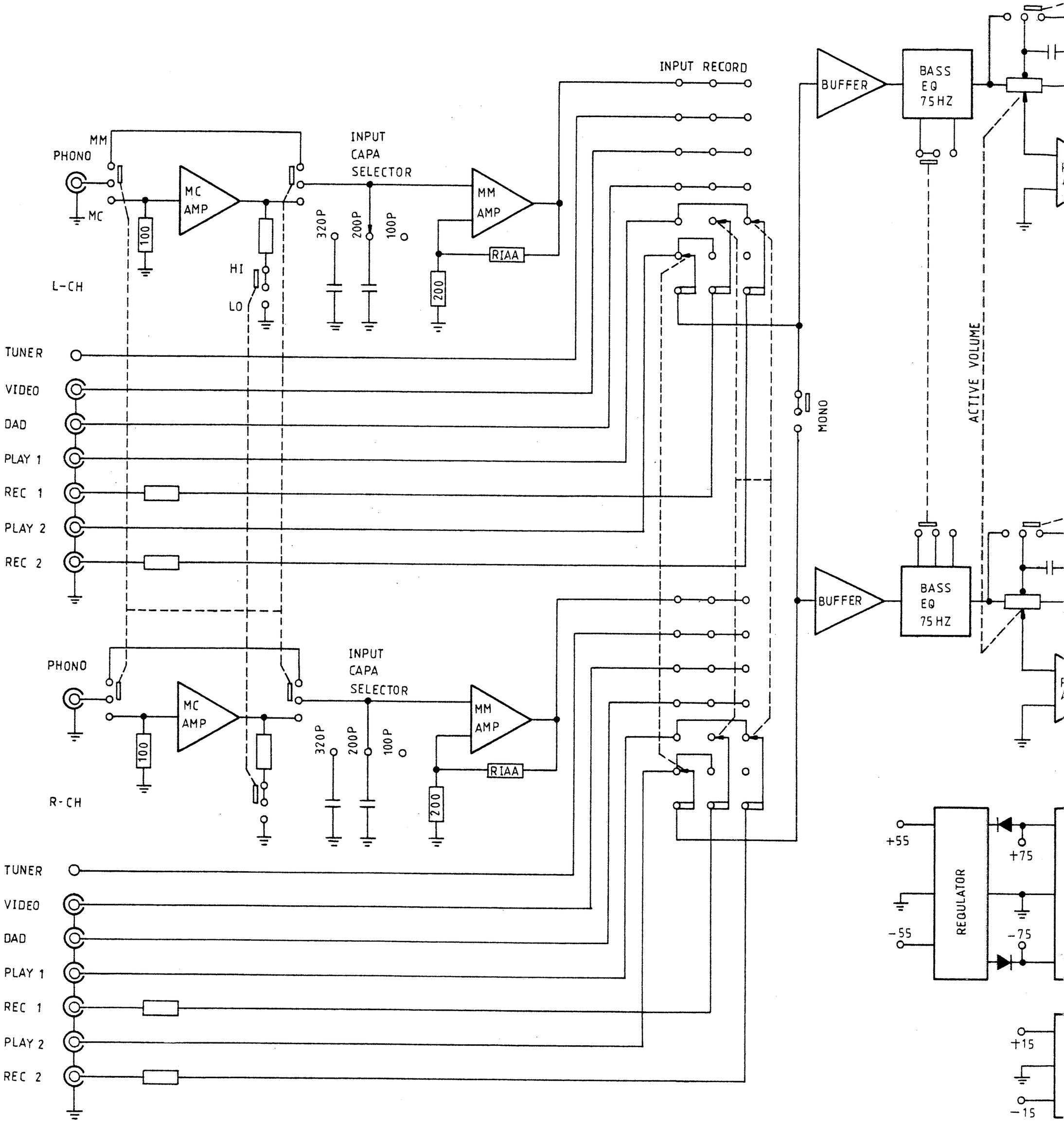
3. AM ALIGNMENT

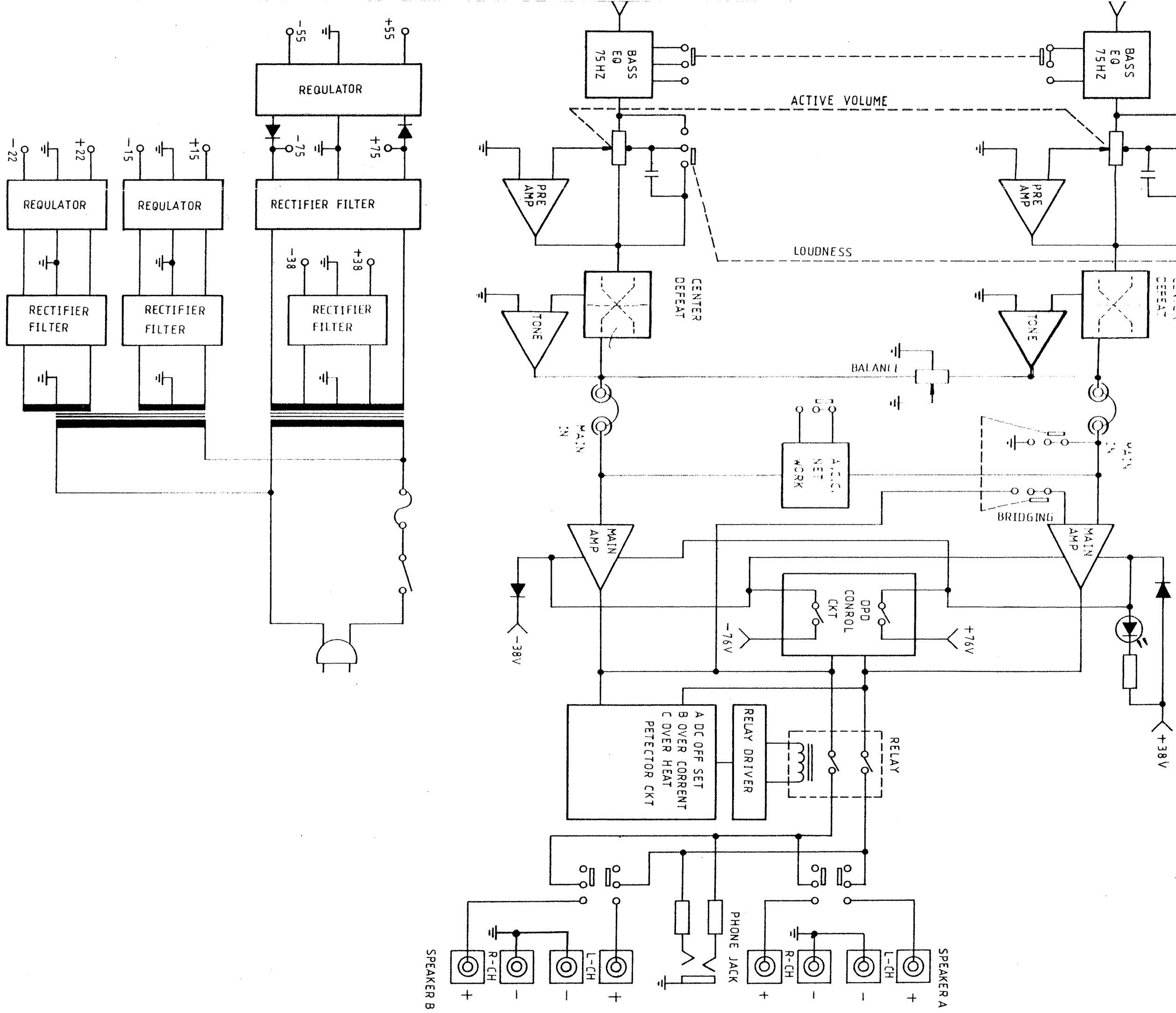
SECTION		AM SG			DIAL SETTING	INDICATOR	ADJUSTEMENT	ADJUST FOR
		CONNECTION	CARRIER	MOD				
AM IF		HOT SIDE OF SG OUTPUT THROUGH 200PF TO AM ANT	450KHz	30% MOD	POINT OF NON-INTERFERENCE	VTVM & OSCILLOSCOPE	IFT403 IFT402	MAXIMUM OUTPUT
AM RANGE	1	HOT SIDE OF SG OUTPUT THROUGH A LOOP ANT RADIATE TO AM LOOP ANT PARALLELY AND KEEP 60cm DISTANCE	600KHz	30% MOD 400Hz	600KHz	DIGITAL VOLTAGE METER	IFT401	PIN TP2 TUNING VOLTAGE 4.5V+-0.1V
	2		1400KHz	30% MOD 400Hz	1400KHz		TC401	PIN TP2 TUNING VOLTAGE 20.5V+-0.2V
	3		REPEAT STEP 1 AND STEP 2					
AM SENS	1		600KHz	30% MOD 400Hz	600KHz	VTVM & OSCILLOSCOPE	IFT404	MAXIMUM OUTPUT
	2		1400KHz		1400KHz		TC402	
	3		REPEAT STEP 1 AND STEP 2					
SIGNAL STRENGTH INDICATOR			1000KHz SG OUTPUT LEVEL 80dB	30% MOD 400Hz	1000KHz	VTVM & OSCILLOSCOPE	VR401	THE 5th LAMP JUST LIGHT

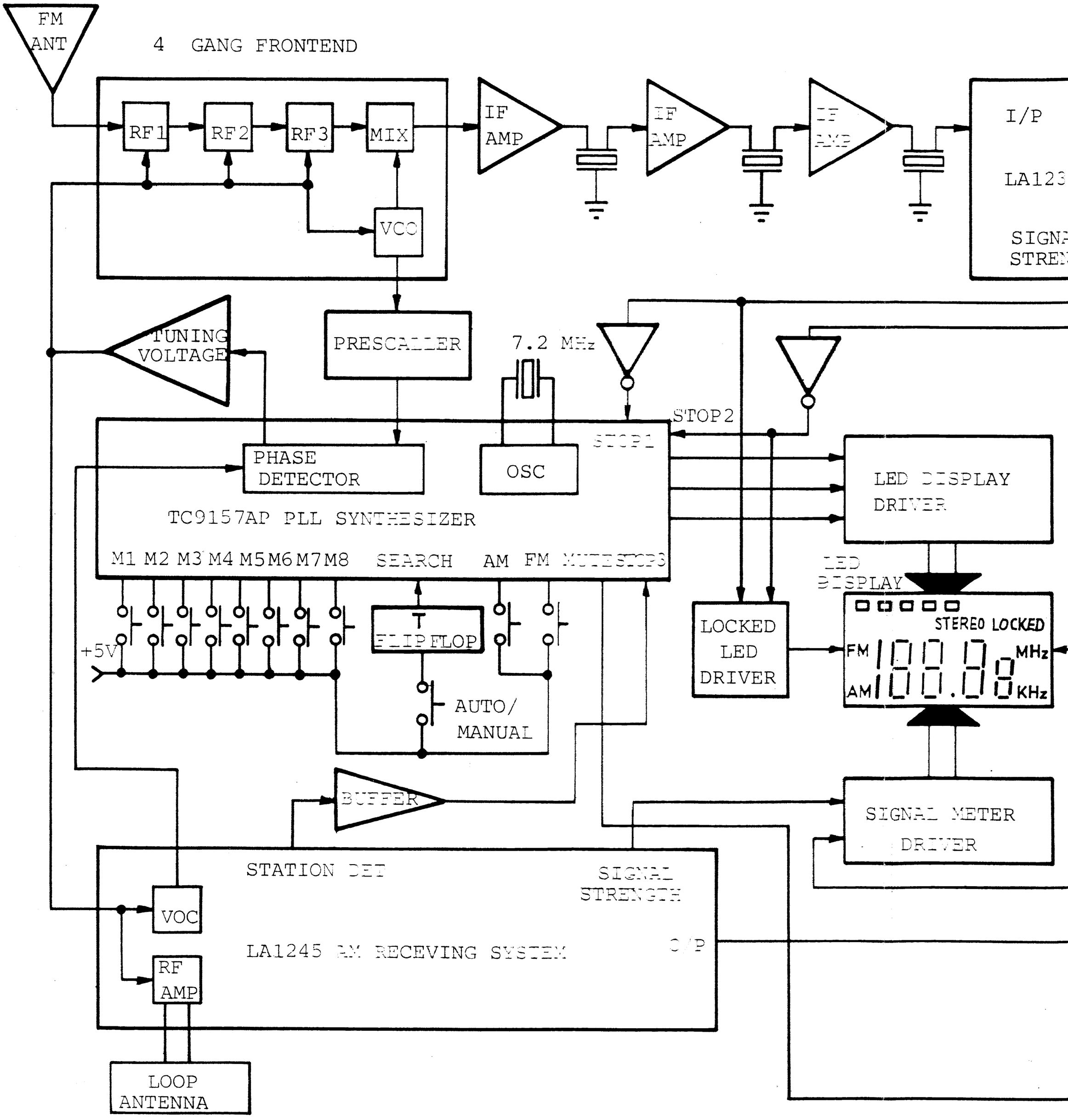
4. AMPLIFIER ALIGNMENT (IDLE CURRENT ADJUSTMENT)

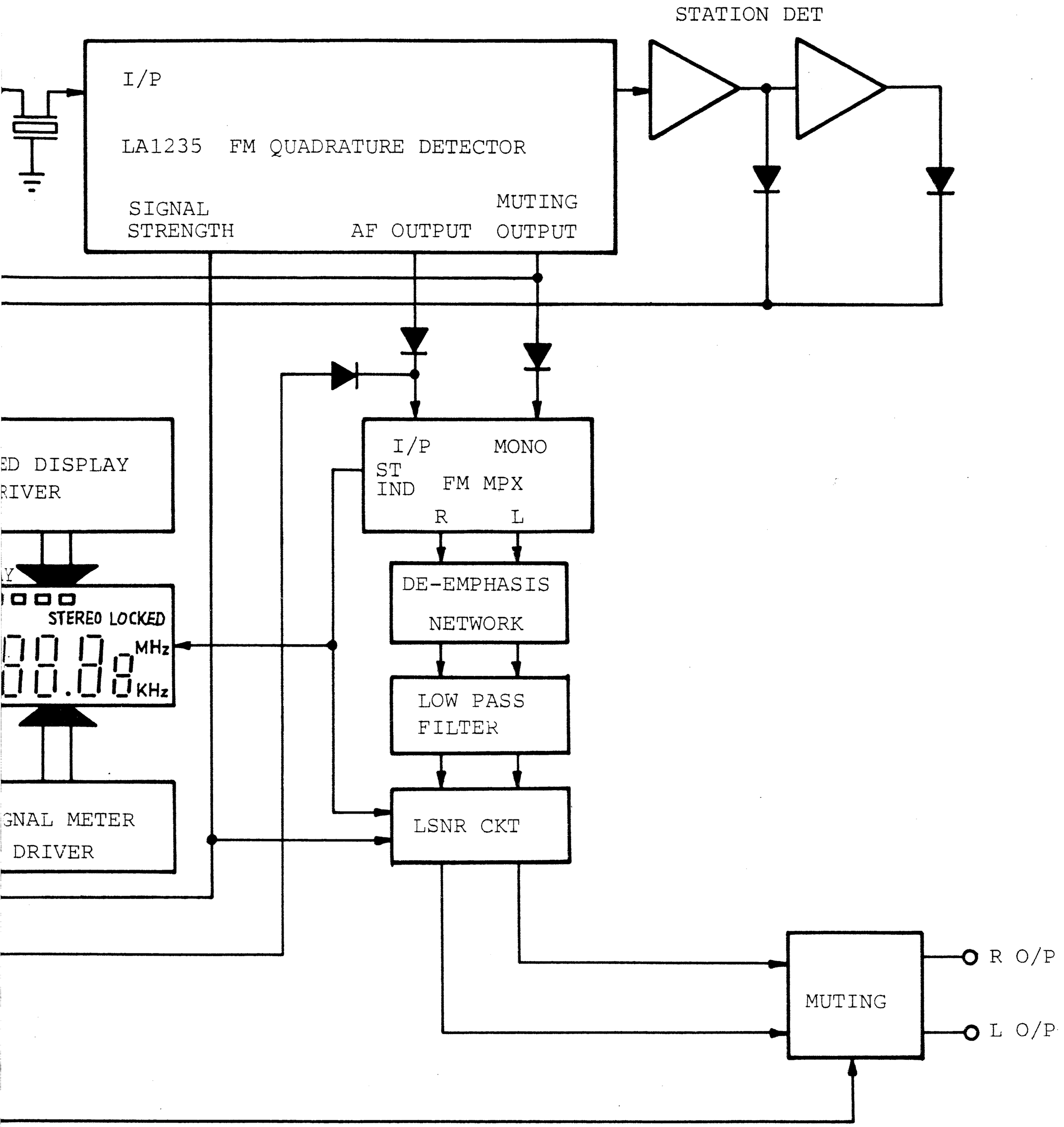
1) Set the volume control at minimum position.
2) Push on POWER for 5 minutes pre-heating.
3) Remove the load from speaker terminals.
4) Connect DC millivolt-meter across R635 (0.1 OHM) for left channel and R636 (0.1 OHM) for right channel. The meter sensitivity should be set at 100mV full scale deflection.
5) If the reading from meter is between 1.5mV and 3.0mV the alignment is completed.
6) If the reading were less than 1.5mV or more than 3.0mV, should be alignment the value of VR601 (left channel) VR602 (right channel) till the reading is between 1.5mV and 3.0mV.

BLOCK DIAGRAM

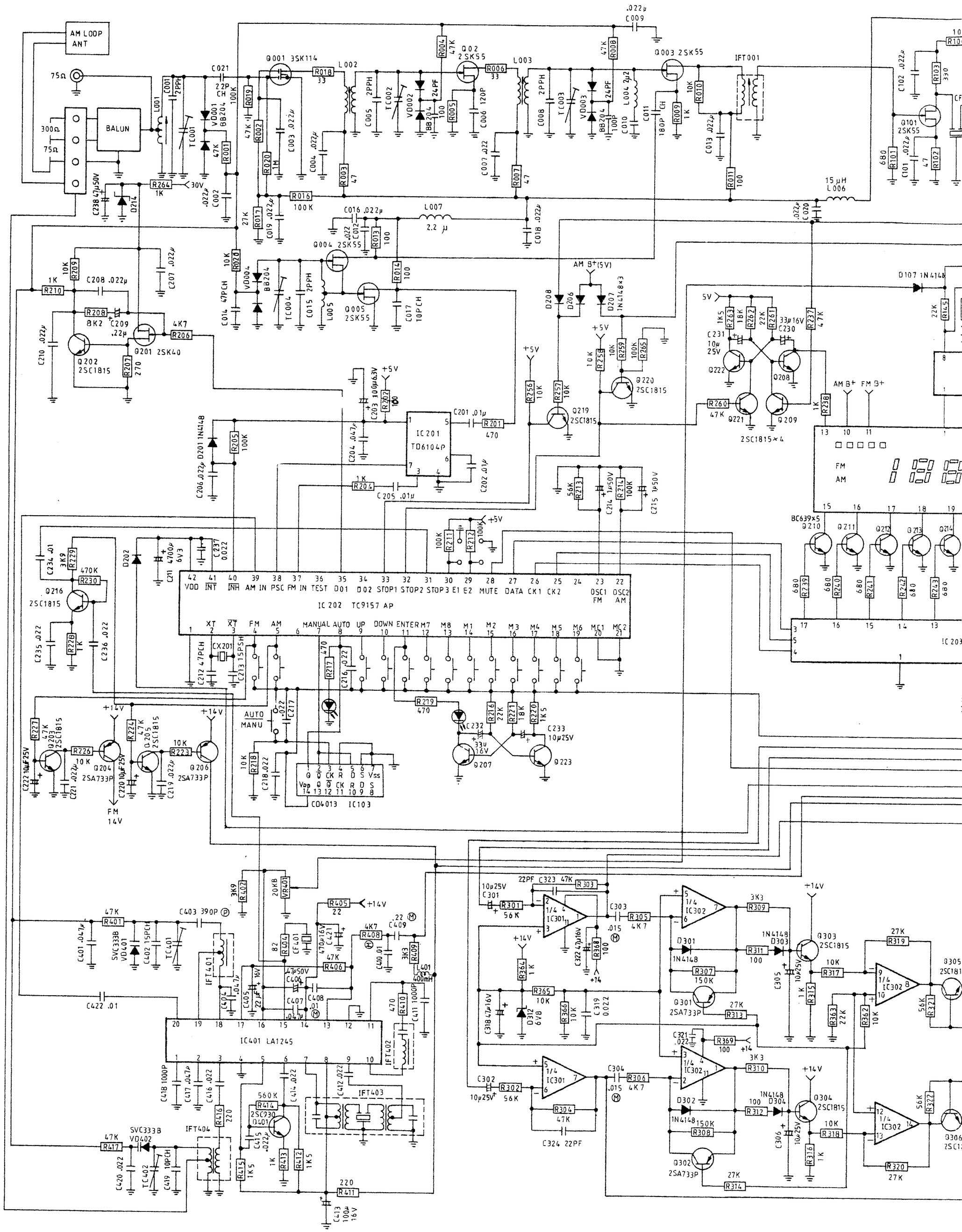


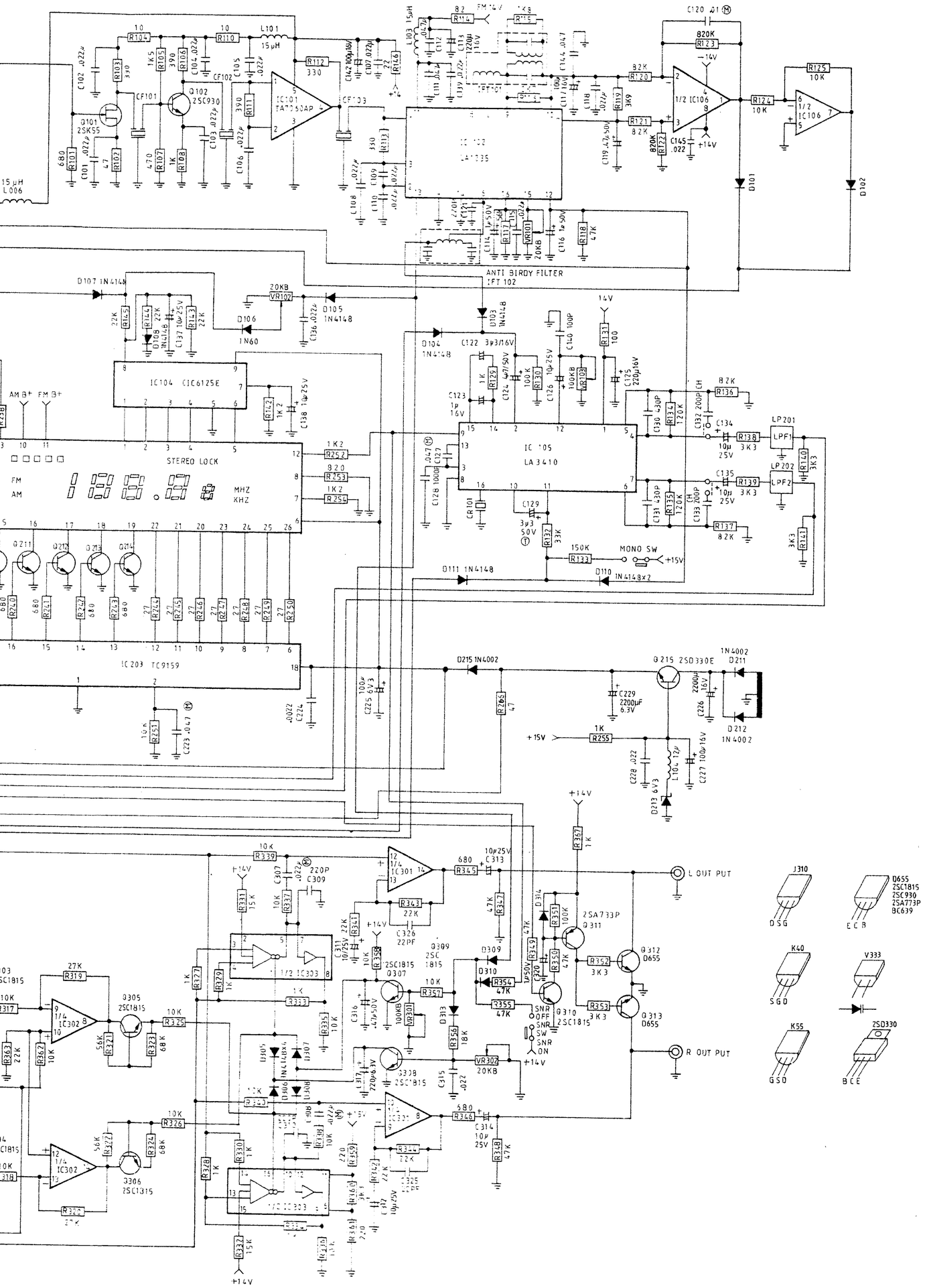

-55
+55
REQULATOR
-22
+22
-15
+15
-75
+75
REQULATOR
REQULATOR
RECTIFIER FILTER
-38
+38
RECTIFIER FILTER
RECTIFIER FILTER
RECTIFIER FILTER
BASS EQ 75HZ
BASS EQ 75HZ
ACTIVE VOLUME
PRE AMP
PRE AMP
LOUDNESS
TONE
CENTER DEFEAT
TONE
CENTER DEFEAT
BALANCE
MAIN IN
MAIN IN
BRIDGING
MAIN AMP
MAIN AMP
-38V
-76V
+76V
+38V
A DC OFF SET
B OVER CORRENT
C OVER HEAT
PETECTOR CKT
RELAY DRIVER
RELAY
PHONE JACK
SPEAKER B
SPEAKER A
R-CH
L-CH
R-CH
L-CH
+
-
-
+
+
-
-
+

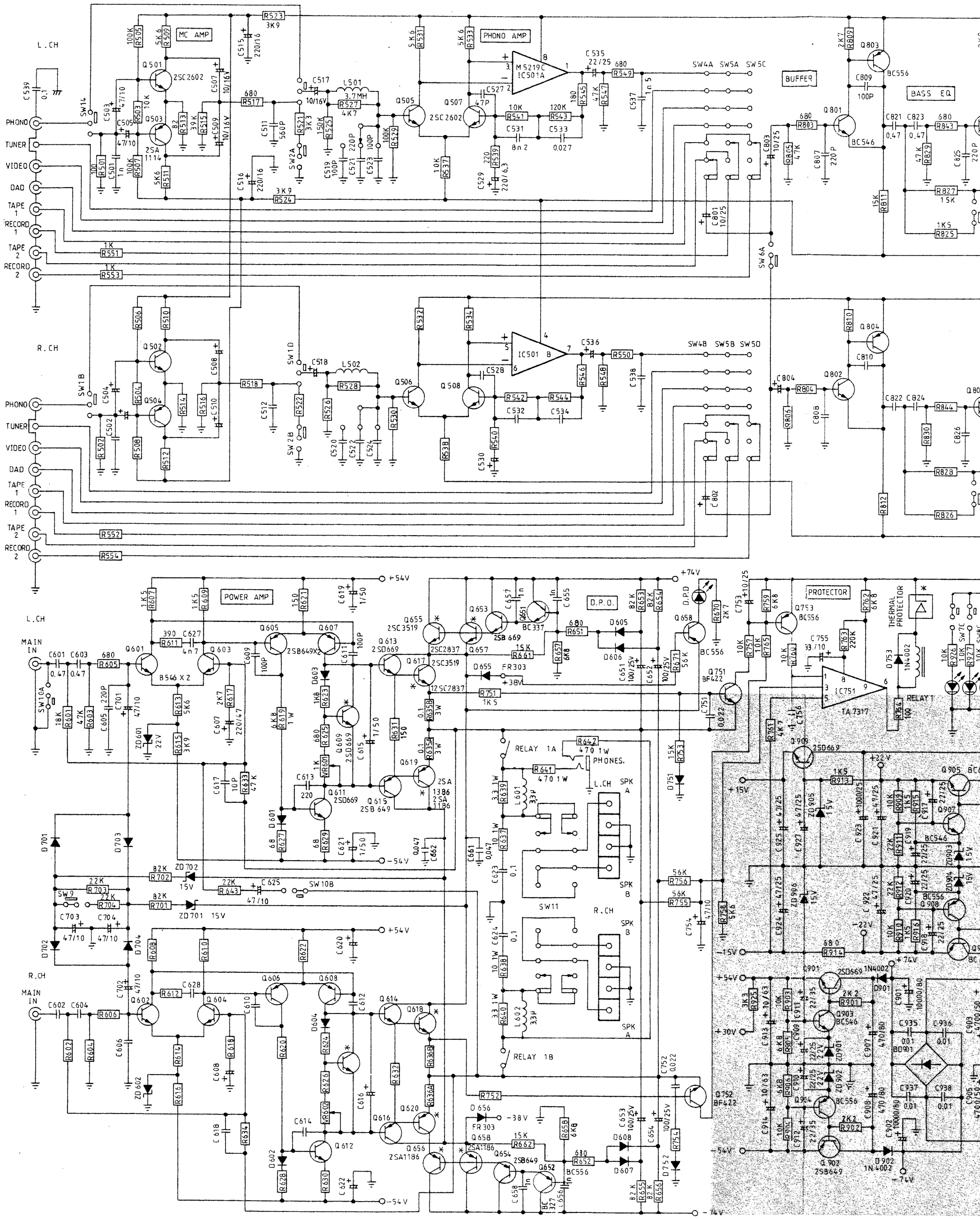

FM
ANT
4 GANG FRONTEND
RF1
RF2
RF3
MIX
VCO
IF
AMP
IF
AMP
I/P
LA123
SIGNA
STREN
TUNING
VOLTAGE
PRESCALLER
7.2 MHz
STOP2
STOP1
PHASE
DETECTOR
OSC
LED DISPLAY
DRIVER
TC9157AP PLL SYNTHESIZER
M1 M2 M3 M4 M5 M6 M7 M8 SEARCH AM FM MUTE STOP3
LED
DISPLAY
+5V
T
FLIP FLOP
LOCKED
LED
DRIVER
STEREO LOCKED
FM
MHz
AM
KHz
AUTO/
MANUAL
BUFFER
SIGNAL METER
DRIVER
STATION DET
SIGNAL
STRENGTH
VOC
LA1245 AM RECEVING SYSTEM
O/P
RF
AMP
LOOP
ANTENNA

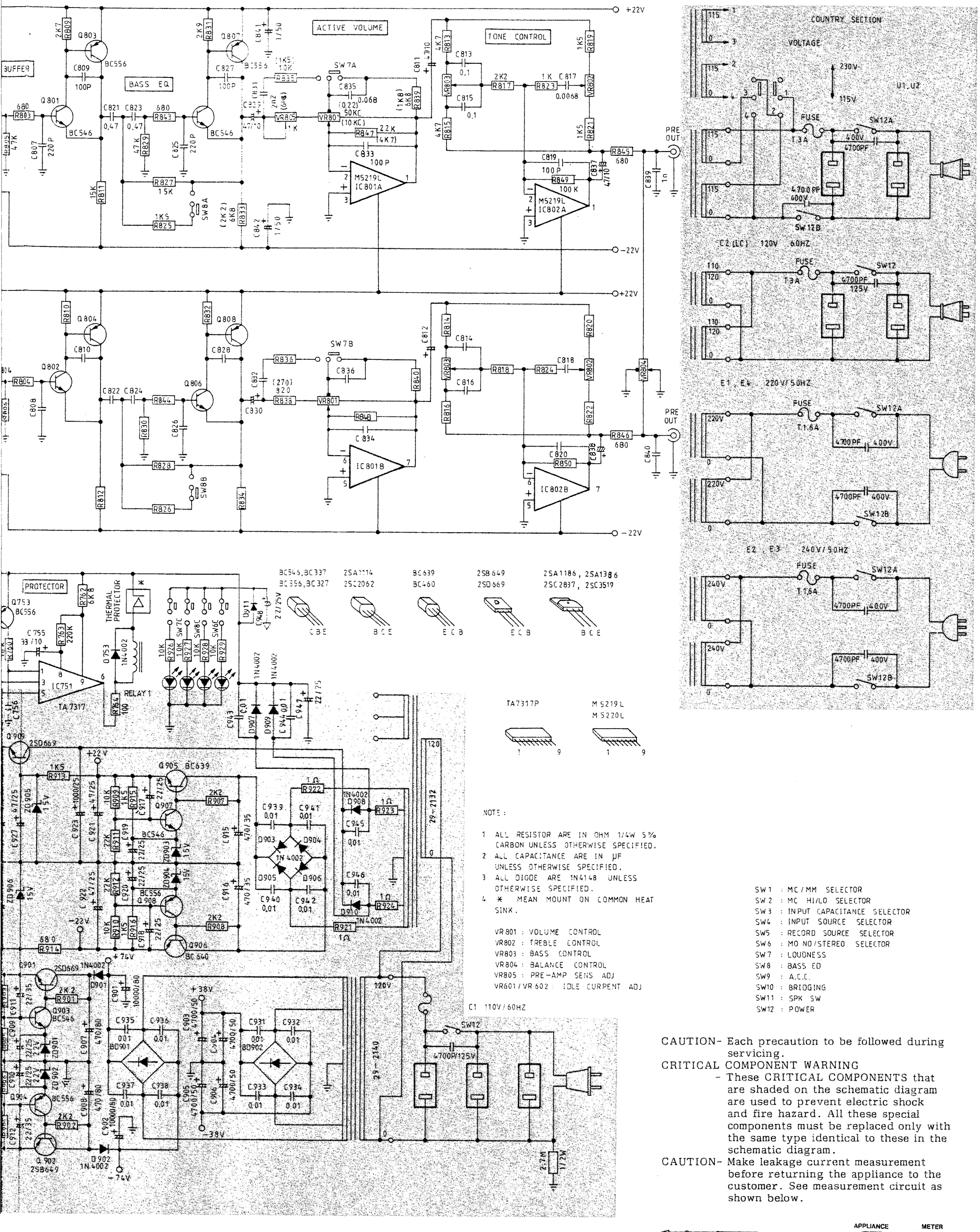

STATION DET
I/P
LA1235 FM QUADRATURE DETECTOR
SIGNAL STRENGTH
AF OUTPUT
MUTING OUTPUT
ED DISPLAY
RIVER
STEREO LOCKED
MHz
KHz
GNAL METER
DRIVER
I/P
MONO
ST IND
FM MPX
R
L
DE-EMPHASIS
NETWORK
LOW PASS
FILTER
LSNR CKT
MUTING
R O/P
L O/P

CIRCUIT DIAGRAM




IC101 TA7650AP
IC102 LA1835
ANTI BIRDY FILTER
IFT 102
IC104 CIC6125E
STEREO LOCK
FM
AM
MHZ
KHZ
IC105 LA3410
MONO SW
IC203 TC9159
L OUT PUT
R OUT PUT
SNR OFF
SNR SW
SNR ON
2SA733P
2SC1815
2SC930
2SD330E
1N4002
1N4148
J310
D655
2SC1815
2SC930
2SA733P
BC639
K40
V333
K55
2SD330


L.CH
R.CH
PHONO
TUNER
VIDEO
DAD
TAPE 1
RECORD 1
TAPE 2
RECORD 2
MC AMP
PHONO AMP
BUFFER
BASS EQ
POWER AMP
PROTECTOR
D.P.D.
MAIN IN
SW1A
SW1B
SW1C
SW1D
SW2A
SW2B
SW4A SW5A SW5C
SW4B SW5B SW5D
SW6A
SW9
SW10A
SW10B
SW11
RELAY 1A
RELAY 1B
PHONES.
SPK A
SPK B
M5219C IC501A
IC501 B
IC751 TA7317
2SC2602
2SA 1114
2SC3519
2SB649X2
2SD669
2SB649
2SA 1186
BC546
BC556
BC337
BF422
FR303
THERMAL PROTECTOR
+74V
+54V
+38V
+30V
+22V
+15V
-74V
-54V
-38V
-22V
-15V



NOTE :

1 ALL RESISTOR ARE IN OHM 1/4W 5% CARBON UNLESS OTHERWISE SPECIFIED.
2 ALL CAPACITANCE ARE IN μF UNLESS OTHERWISE SPECIFIED.
3 ALL DIODE ARE 1N4148 UNLESS OTHERWISE SPECIFIED.
4 * MEAN MOUNT ON COMMON HEAT SINK.

VR801 : VOLUME CONTROL
VR802 : TREBLE CONTROL
VR803 : BASS CONTROL
VR804 : BALANCE CONTROL
VR805 : PRE-AMP SENS ADJ
VR601/VR602 : IDLE CURPENT ADJ

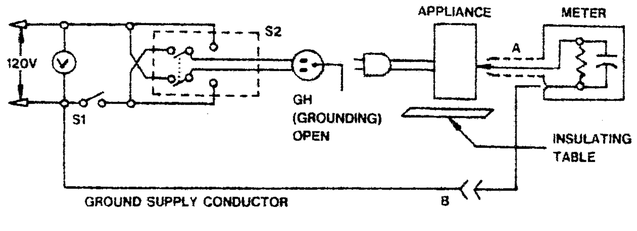
C1 110V/60HZ

SW1 : MC/MM SELECTOR
SW2 : MC HI/LO SELECTOR
SW3 : INPUT CAPACITANCE SELECTOR
SW4 : INPUT SOURCE SELECTOR
SW5 : RECORD SOURCE SELECTOR
SW6 : MO NO/STEREO SELECTOR
SW7 : LOUDNESS
SW8 : BASS EQ
SW9 : A.C.C.
SW10 : BRIDGING
SW11 : SPK SW
SW12 : POWER

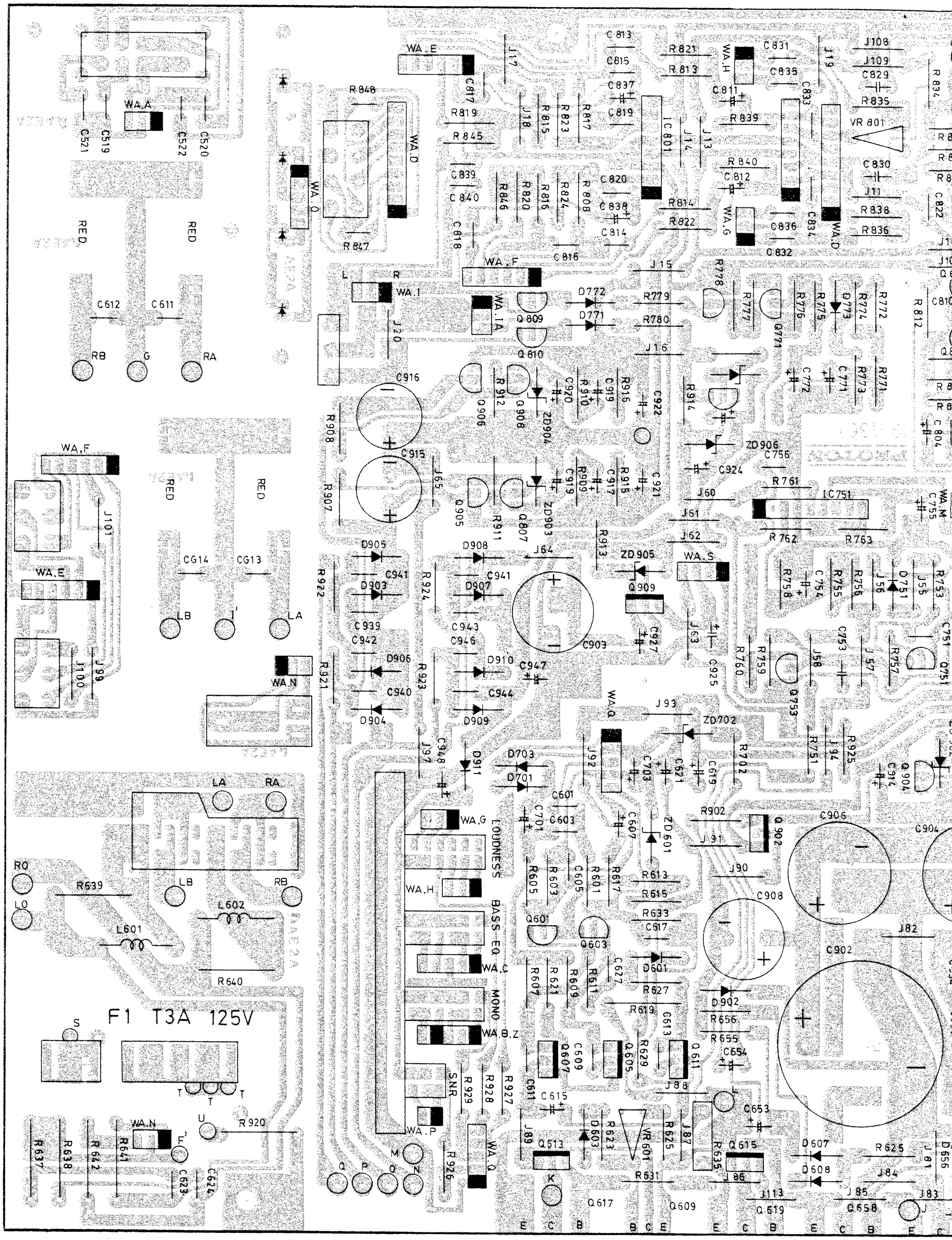
CAUTION- Each precaution to be followed during servicing.

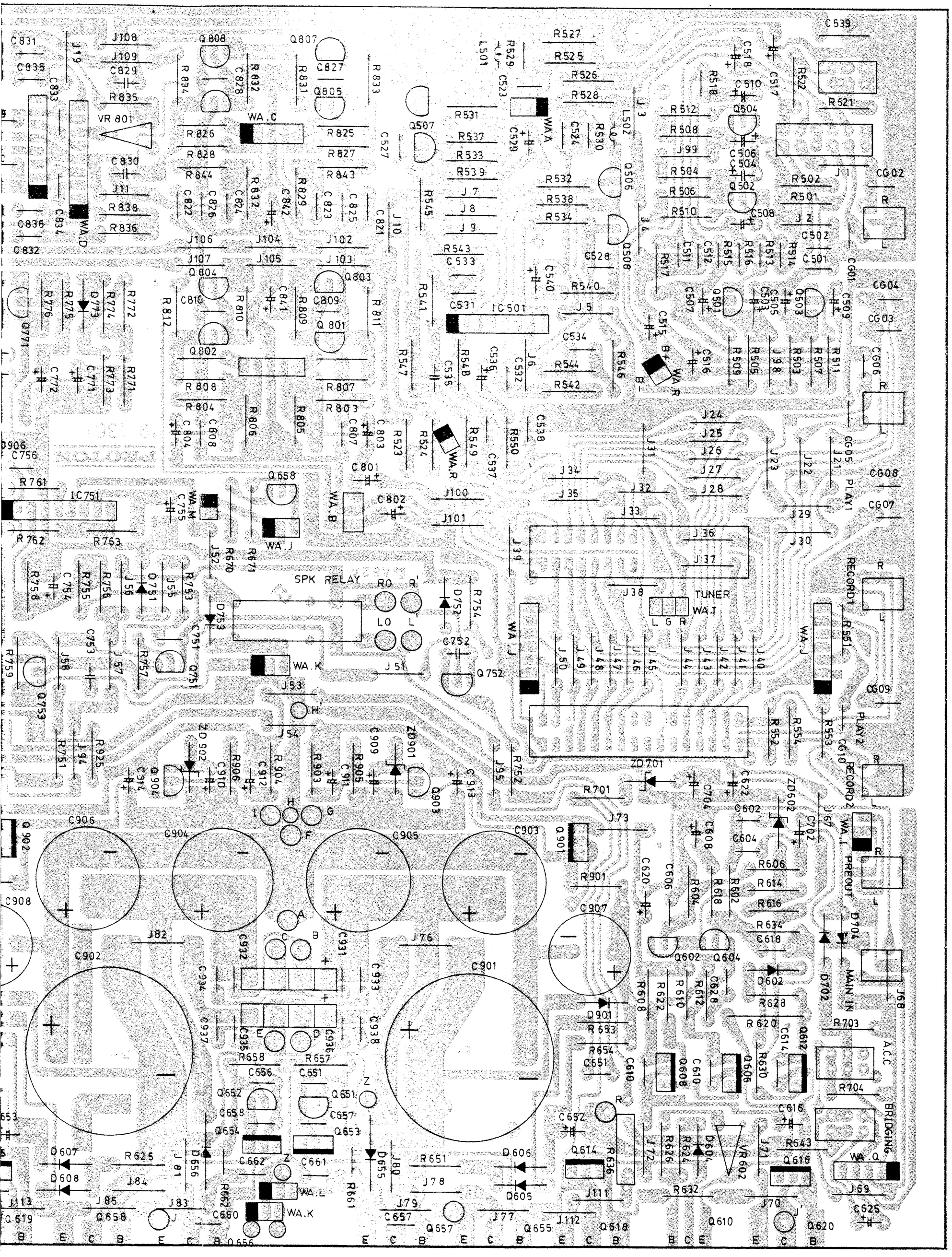
CRITICAL COMPONENT WARNING
- These CRITICAL COMPONENTS that are shaded on the schematic diagram are used to prevent electric shock and fire hazard. All these special components must be replaced only with the same type identical to these in the schematic diagram.

CAUTION- Make leakage current measurement before returning the appliance to the customer. See measurement circuit as shown below.



PCB PARTS LOCATION (TOP VIEW)




C539
R522
R521
C517
C518
C510
R518
Q504
R512
R508
J99
C506
C504
R504
Q502
R506
R510
C508
J 1
CG02
R
R502
R501
J 2
C502
C501
CG01
C509
Q503
C505
C503
Q501
C507
C515
C516
WA.R
B+
B-
CG04
CG03
CG06
R
R509
R505
J98
R503
R507
R511
J 3
L502
J 4
R527
R525
R526
R528
C524
R530
R531
R537
R533
R539
J 7
J 8
J 9
R543
C533
C531
IC501
C534
R544
R542
R546
C536
C535
R548
R547
R541
R545
Q507
Q506
Q508
R532
R538
R534
C528
C540
R540
J 5
J 6
C532
C538
R550
R549
C537
R524
WA.R
C523
R529
L501
C529
WA.A
C527
R833
C821
J 10
C831
C835
C833
J19
J108
J109
C829
R835
VR 801
C830
J11
R838
R836
C832
C836
C834
WA.D
Q808
Q807
C827
Q805
R831
R832
C828
R834
WA.C
R826
R828
R844
C822
C826
C824
R825
R827
R843
C823
C825
C842
R829
R832
C842
J106
J107
J104
J105
J102
J103
Q804
Q803
C810
C809
R810
R809
R811
R812
C841
Q801
Q802
R808
R804
C804
C808
R806
R805
R807
R803
C807
C803
C801
C802
WA.B
J100
J101
Q658
WA.J
R761
IC751
R762
R763
WA.M
C755
J52
R670
R671
R772
R774
D773
R775
R776
C772
C771
R773
R771
Q771
Q906
C756
SPK RELAY
RO
R'
LO
L'
J 51
WA.K
J53
H'
J54
R753
D753
C751
Q751
R757
C753
J 57
J58
R759
Q753
D751
J56
J55
R755
R756
C754
R758
D752
R754
C752
Q752
R752
J95
R925
J94
R751
Q904
C914
R906
C910
ZD 902
R904
C912
C909
R905
C911
R903
C913
ZD901
Q903
H
I
G
F
Q902
C906
C904
C905
C903
Q901
R901
J73
C907
D901
R653
R654
C651
C610
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C620
C606
R604
R618
R602
C608
C604
ZD701
R701
C704
C622
C602
ZD602
J67
C702
WA.J
PREOUT
R
R606
R614
R616
R634
C618
D602
R628
R620
D704
D702
MAIN IN
J68
R703
A.C.C
R704
C614
Q612
Q608
Q606
C610
R630
C616
BRIDGING
WA.Q
R643
Q616
J69
C625
J71
J72
R626
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R612
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C902
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C937
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C935
C936
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D
C
B
A
Z
R'
J76
C901
R658
R657
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C651
Q652
Q651
C658
C657
Q654
Q653
C662
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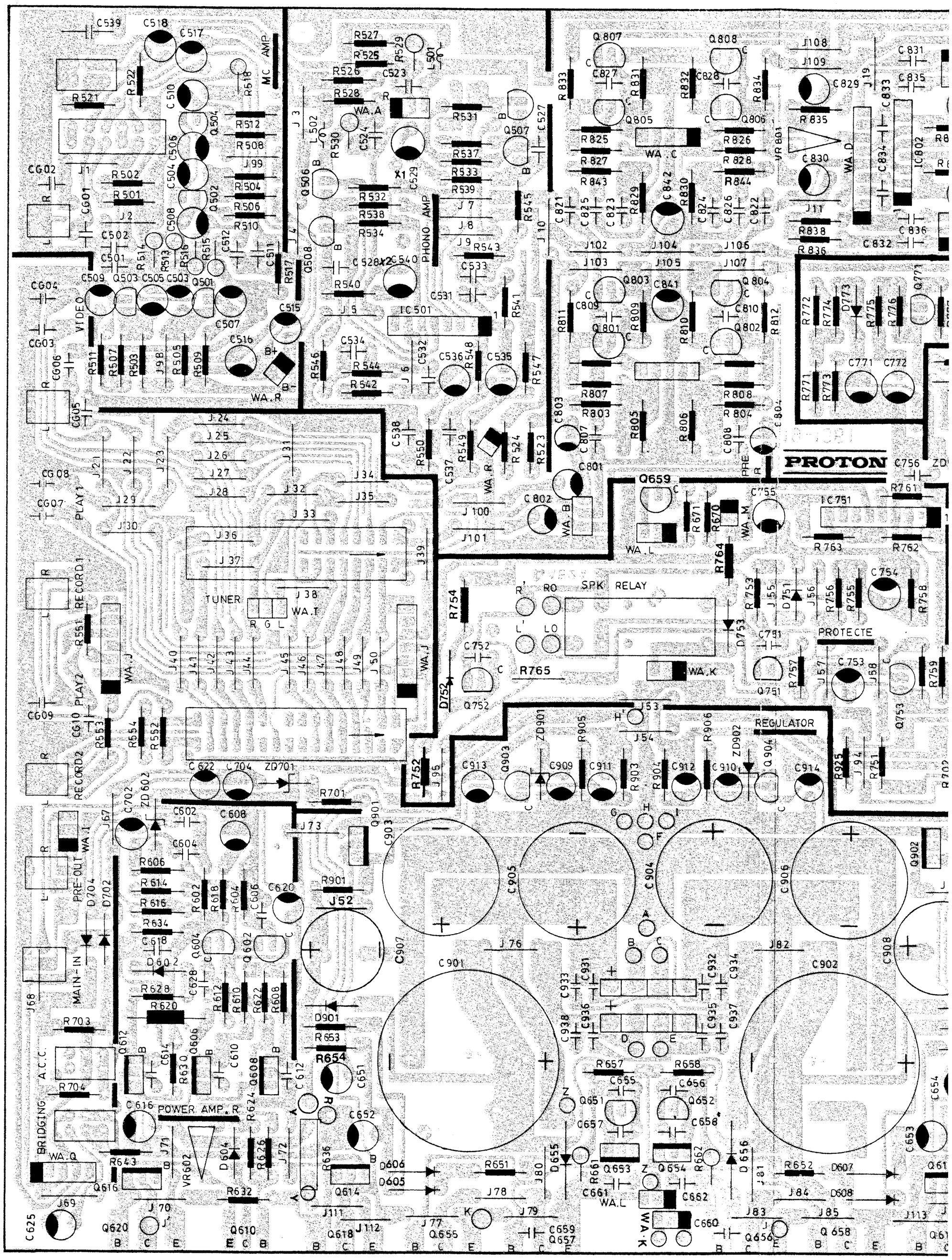
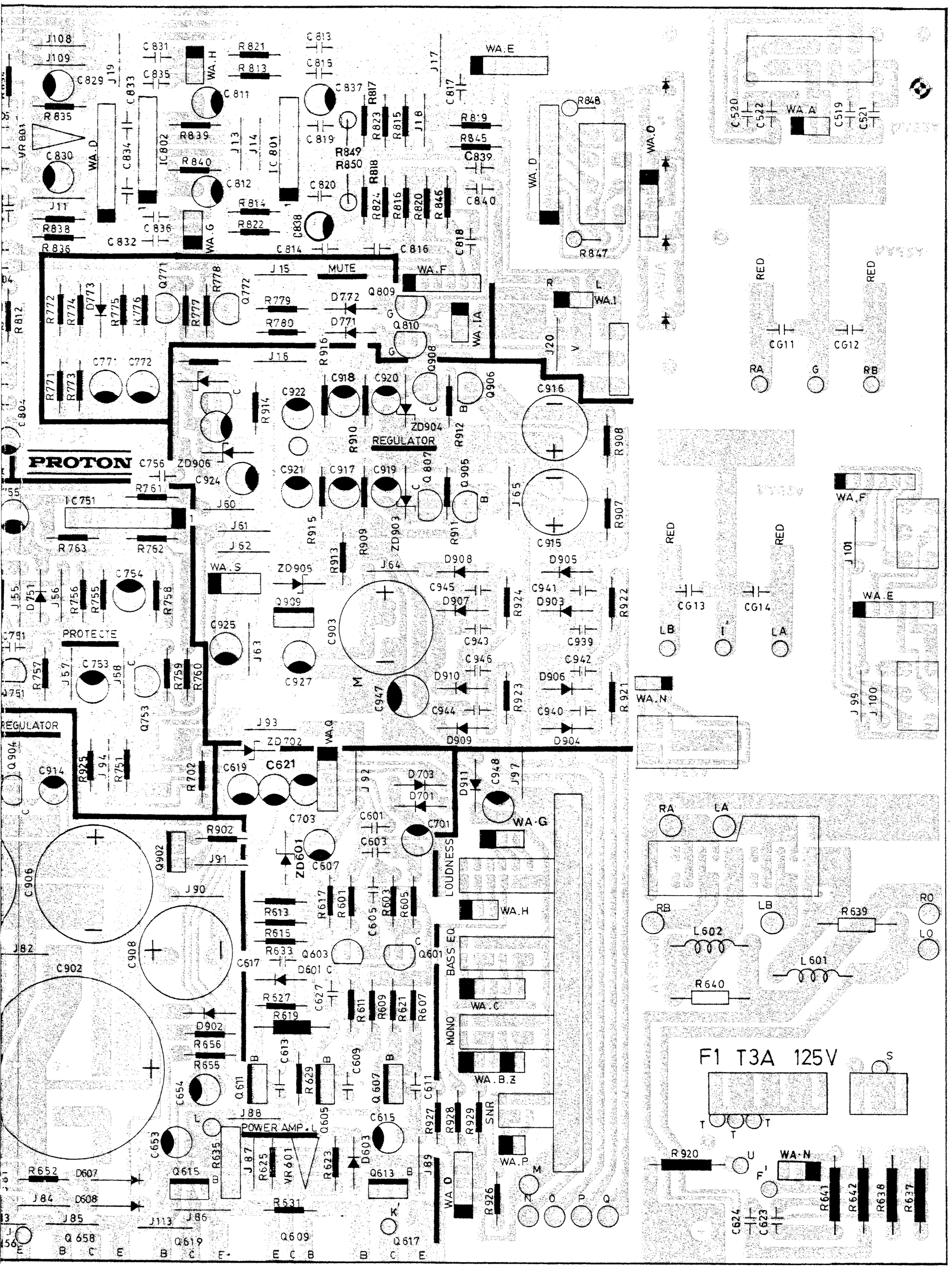
WA.L
WA.K
C660
R662
R661
D655
J80
R651
J78
J79
C657
Q657
J77
Q655
J112
Q618
J111
D606
D605
Q614
R636
D607
D608
R625
J84
J85
J83
J81
D656
Q658
Q656
J113
Q619
J24
J25
J26
J27
J28
J23
J22
J21
CG05
PLAY1
CG08
CG07
J29
J30
J31
J34
J35
J32
J33
J36
J37
J39
J38
TUNER
WA.T
L G R
J41
J42
J43
J44
J40
J45
J46
J47
J48
J49
J50
WA.J
RECORD1
R551
R
CG09
PLAY2
R553
R554
R552
CG10
RECORD2
R
WA.J
J67
R
PREOUT
J68
MAIN IN
J69
BRIDGING
WA.Q
J70
J71

PCB PARTS LOCATION (BOTTOM VIEW)




PROTON
REGULATOR
PROTECTE
POWER AMP. L
LOUDNESS
BASS EQ
MONO
SNR
RED
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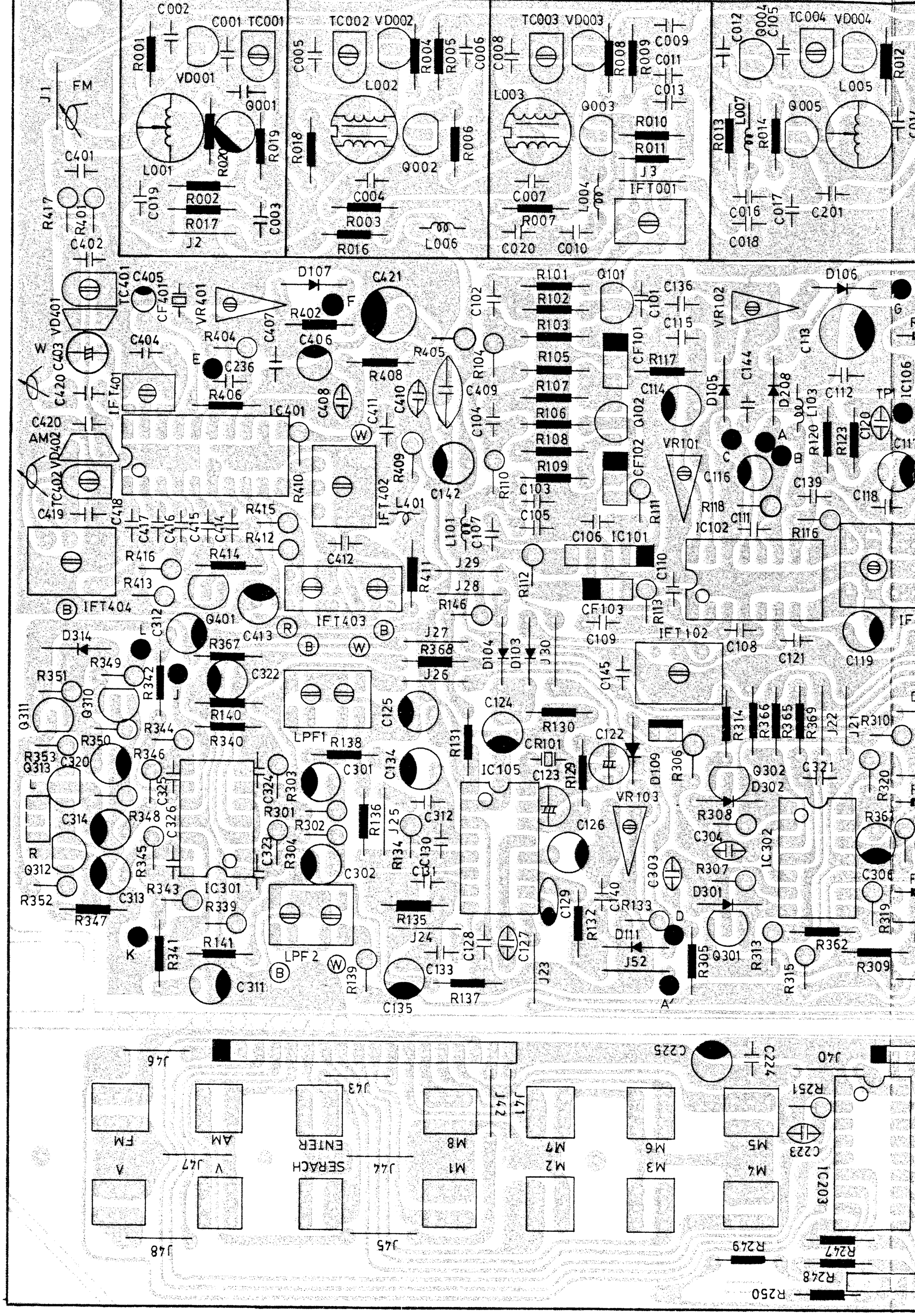
WA.A
WA.B.Z
WA.C
WA.D
WA.E
WA.F
WA.G
WA.H
WA.I
WA.IA
WA.J
WA.N
WA.O
WA.P
WA.Q
MUTE
IC801
IC802
IC751
ZD601
ZD702
ZD903
ZD904
ZD905
ZD906
VR601
VR803
REGULATOR

PCB PARTS LOCATION (TOP VIEW)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TC002 VD002
TC003 VD003
TC004 VD004
VD001
L001
L002
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TC401
C405
CF401
VR401
VD401
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IFT404
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IC102
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CR101
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CF102
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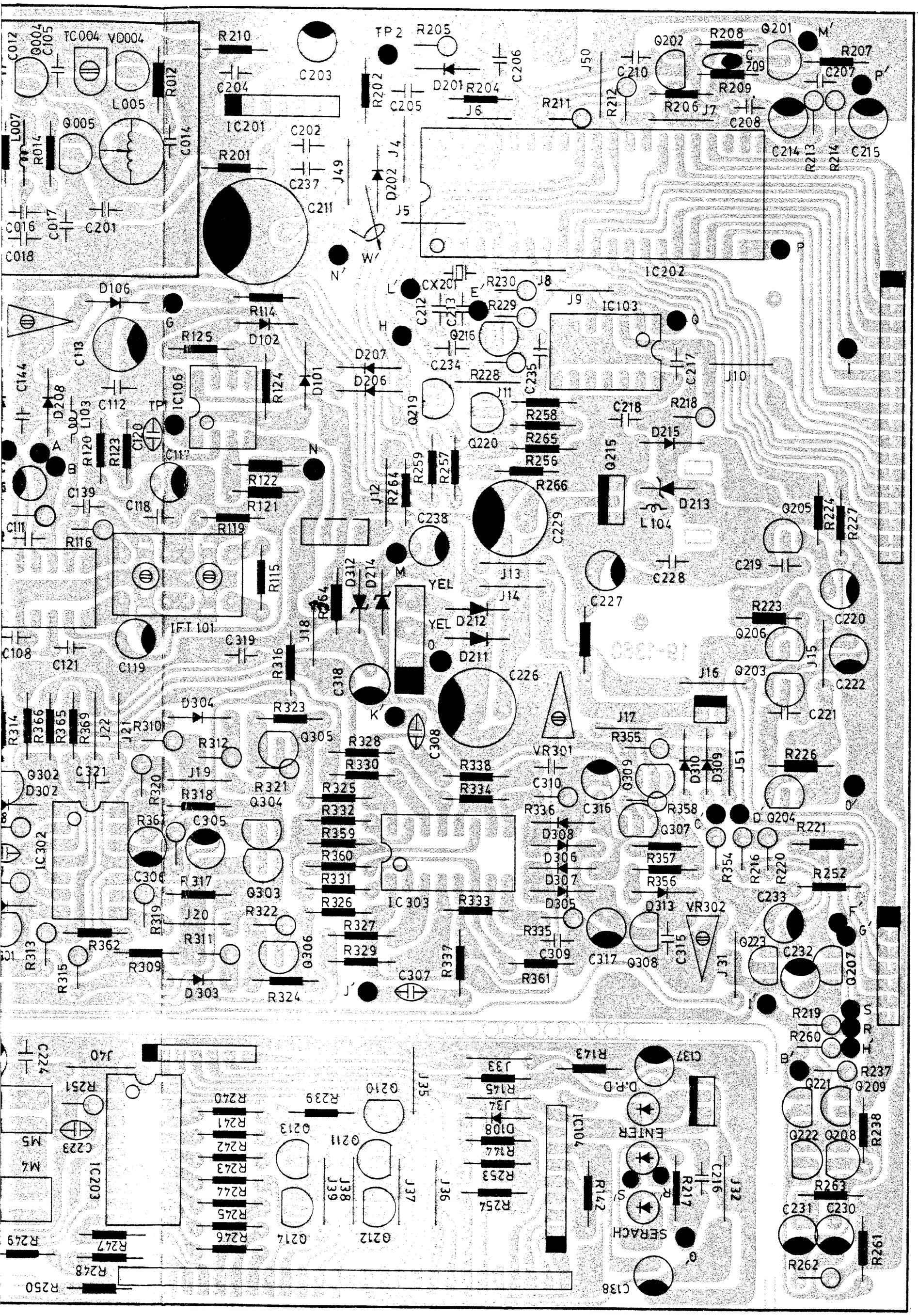
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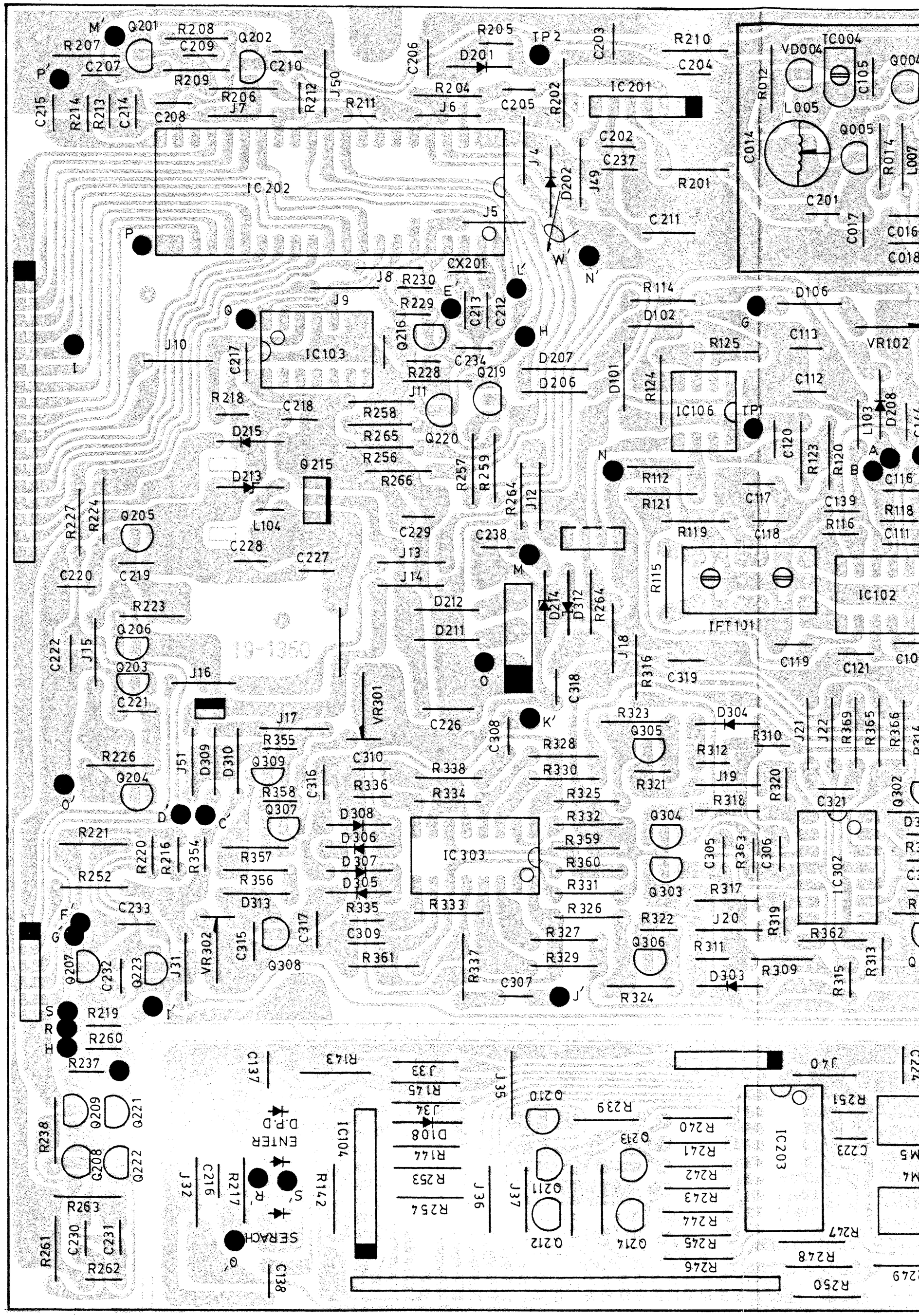
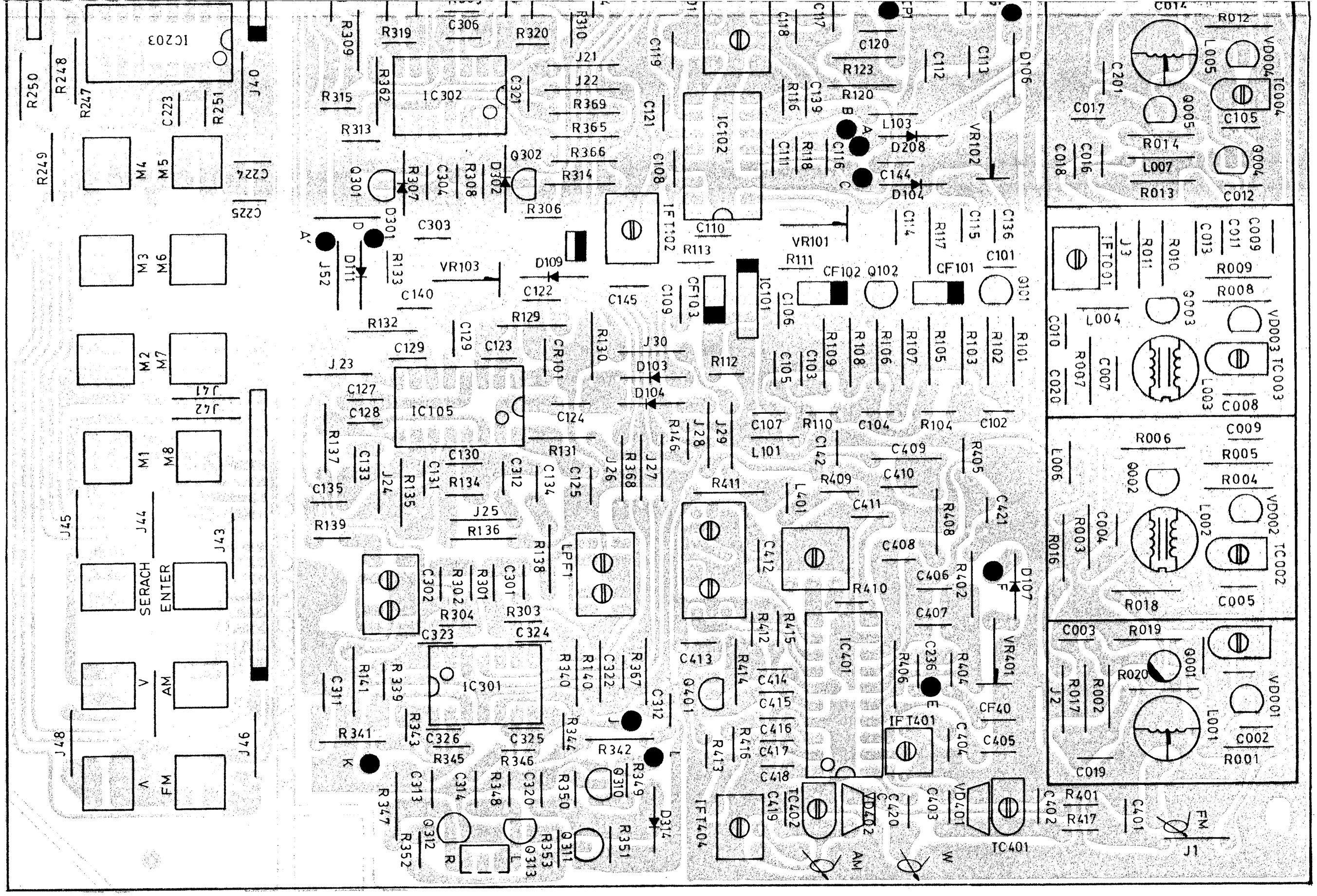
M5

PCB PARTS LOCATION (BOTTOM VIEW)

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IFT404
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CF102
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CF40
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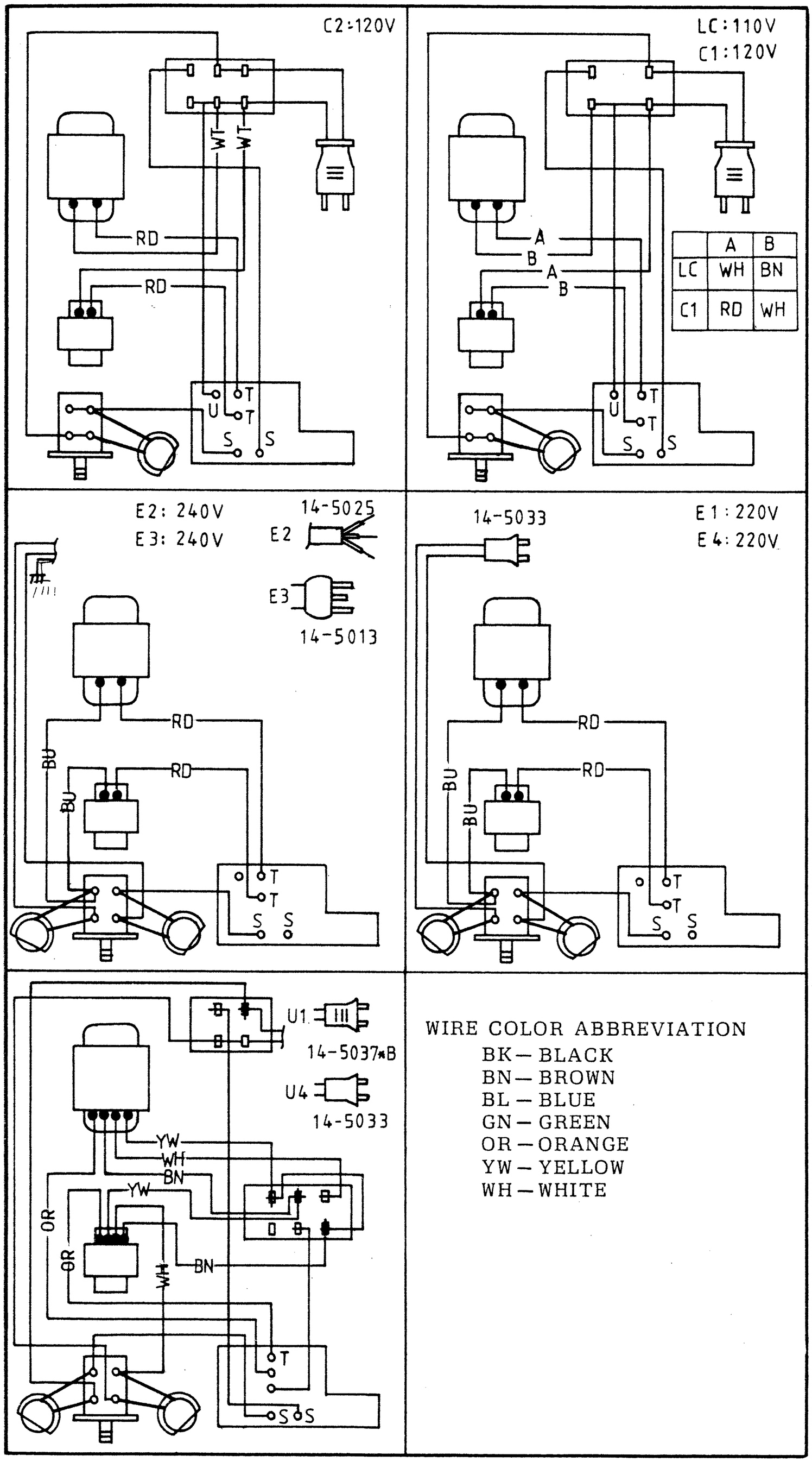
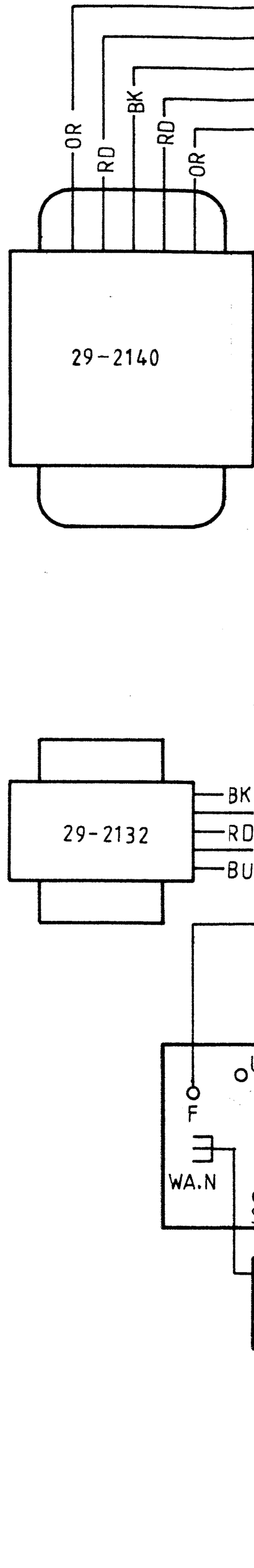
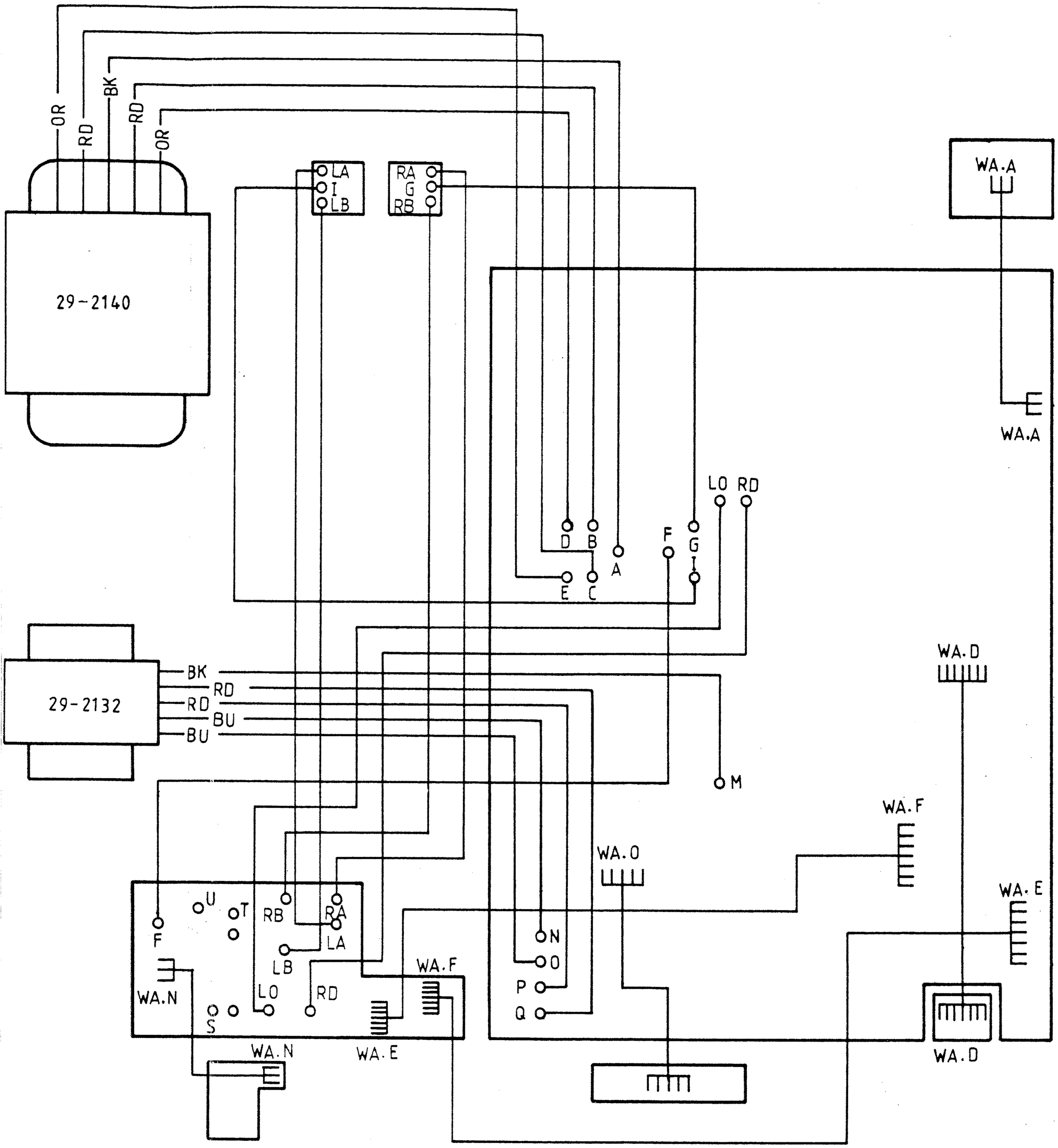
WIRING DIAGRAM






OR
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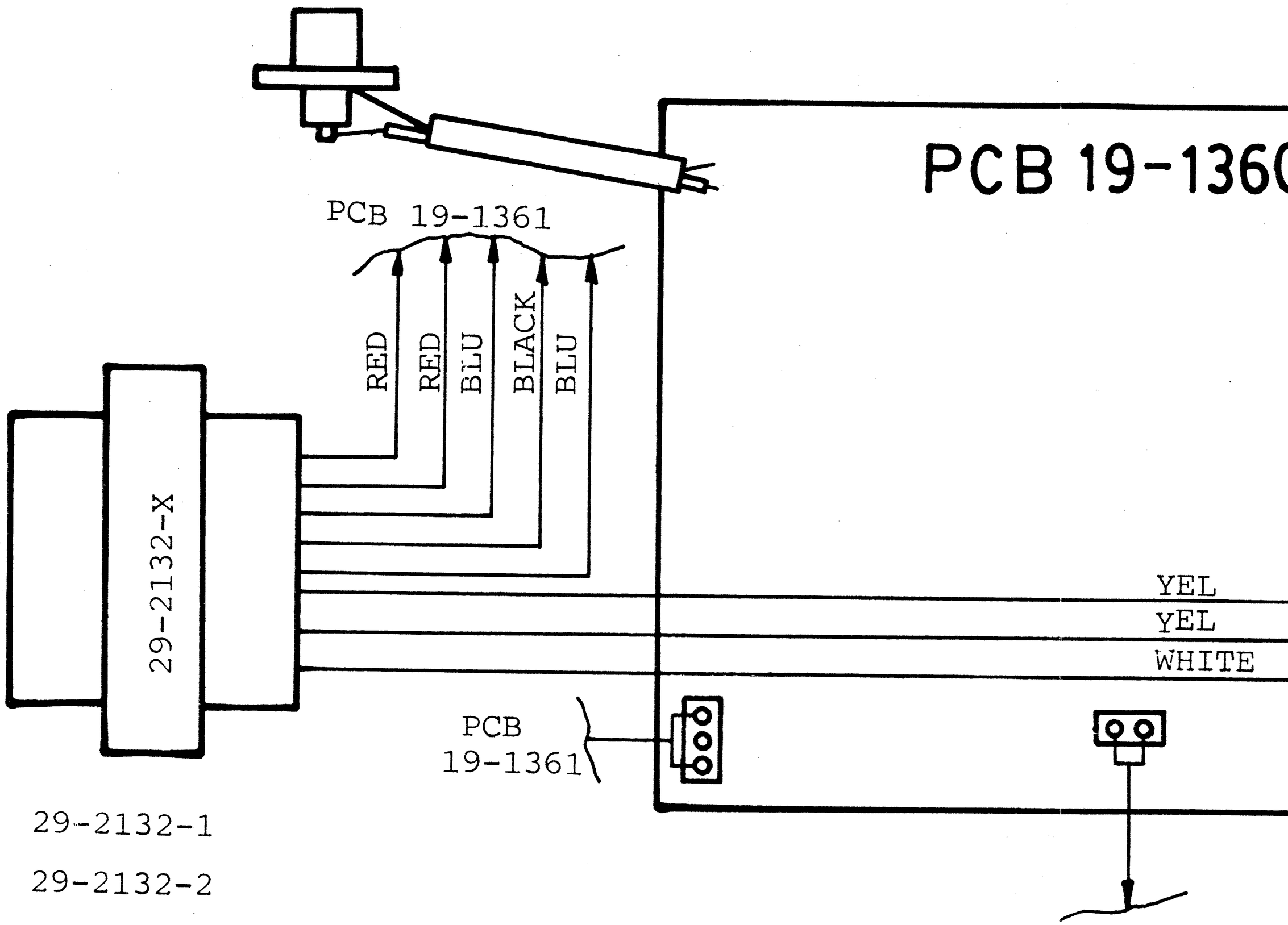
F CONNECTOR FM 75 OHM



29-2132-1

29-2132-2

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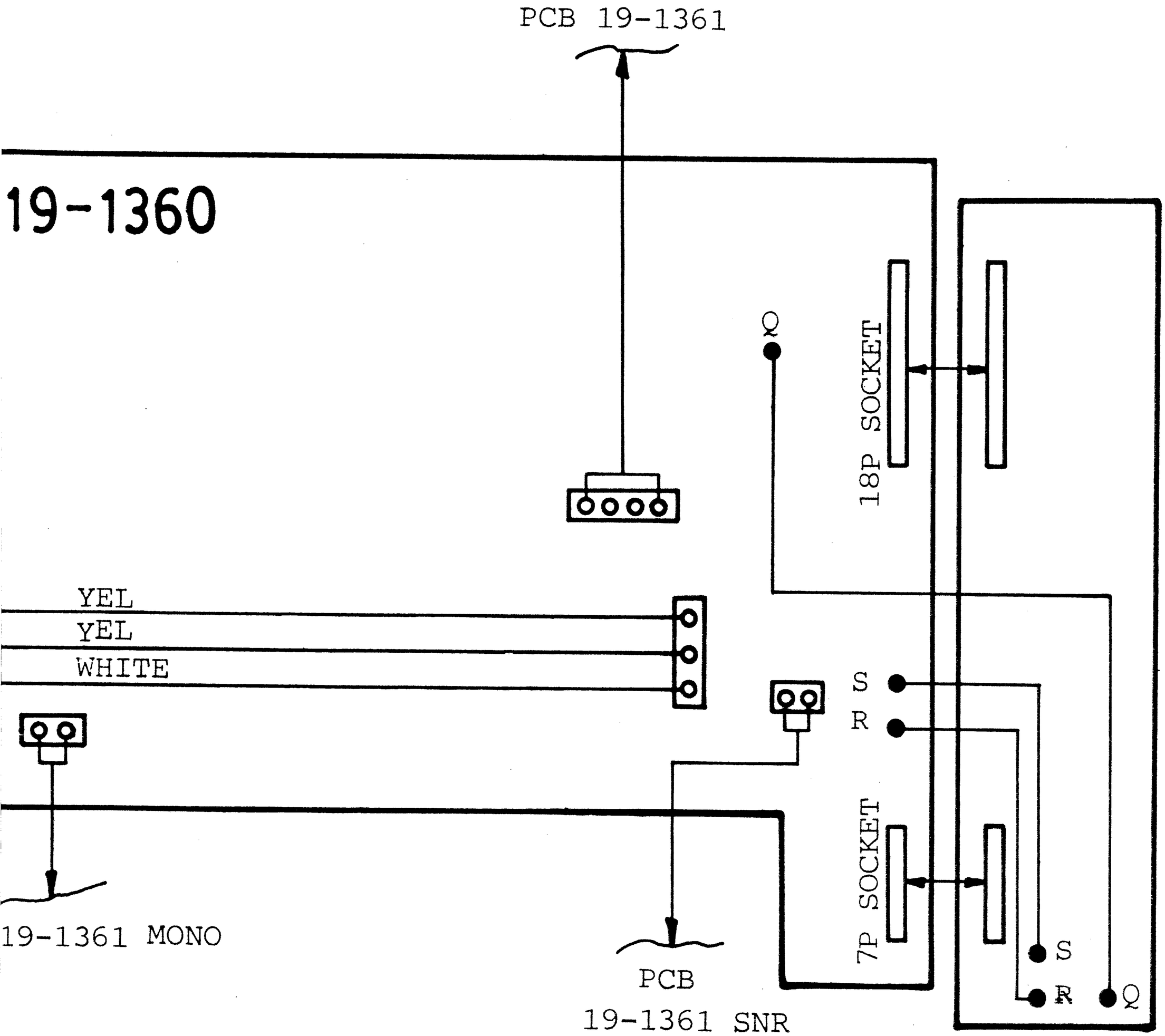
29-2132-4

29-2132-5

29-2132-7


PCB 19-1361
19-1360
YEL
YEL
WHITE
19-1361 MONO
PCB
19-1361 SNR
Q
18P SOCKET
S
R
7P SOCKET
S
R
Q

ELECTRICAL PARTS LIST

REF.NO.	PART NO.	DESCRIPTION
RESISTORS (All are carbon film 1/4W +-5% unless noted otherwise)		
R101	16-1/4CA681J	RES CA 680 OHM 1/4W
R102	16-1/4CA470J	RES CA 47 OHM 1/4W
R103	16-1/4CA331J	RES CA 330 OHM 1/4W
R104	16-1/4CR100J	RES CR 10 OHM 1/4W
R105	16-1/4CA152J	RES CA 1K5 OHM 1/4W
R106	16-1/4CA391J	RES CA 390 OHM 1/4W
R107	16-1/4CA471J	RES CA 470 OHM 1/4W
R108	16-1/4CA102J	RES CA 1K OHM 1/4W
R110	16-1/4CL100J	RES CL SM 10 OHM 1/4W
R111	16-1/4CR391J	RES CR 390 OHM 1/4W
R112	16-1/4CR331J	RES CR 330 OHM 1/4W
R113	16-1/4CR331J	RES CR 330 OHM 1/4W
R115	16-1/4CR182J	RES CR 1K8 OHM 1/4W
R116	16-1/4CR272J	RES CR 2K7 OHM 1/4W
R117	16-1/4CA563J	RES CA 56K OHM 1/4W
R118	16-1/4CR473J	RES CR 47K OHM 1/4W
R119	16-1/4CA392J	RES CA 3K9 OHM 1/4W
R120	16-1/4CA823J	RES CA 82K OHM 1/4W
R121	16-1/4CA823J	RES CA 82K OHM 1/4W
R122	16-1/4CA824J	RES CA 820K OHM 1/4W
R123	16-1/4CA824J	RES CA 820K OHM 1/4W
R124	16-1/4CA103J	RES CA 10K OHM 1/4W
R125	16-1/4CA103J	RES CA 10K OHM 1/4W
R129	16-1/4CA102J	RES CA 1K OHM 1/4W
R130	16-1/4CA104J	RES CA 100K OHM 1/4W
R131	16-1/4CA101J	RES CA 100 OHM 1/4W
R132	16-1/4CA333J	RES CA 33K OHM 1/4W
R133	16-1/4CL154J	RES CL 150K OHM 1/4W
R134	16-1/4CR124J	RES CR 120K OHM 1/4W
R135	16-1/4CA124J	RES CA 120K OHM 1/4W
R136	16-1/4CA823J	RES CA 82K OHM 1/4W
R137	16-1/4CA823J	RES CA 82K OHM 1/4W
R138	16-1/4CA332J	RES CA 3K3 OHM 1/4W
R139	16-1/4CR332J	RES CR 3K3 OHM 1/4W
R140	16-1/4CA332J	RES CA 3K3 OHM 1/4W
R141	16-1/4CA332J	RES CA 3K3 OHM 1/4W
R142	16-1/4CA122J	RES CA 1K2 OHM 1/4W
R143	16-1/4CA223J	RES CA 22K OHM 1/4W
R144	16-1/4CA223J	RES CA 22K OHM 1/4W
R145	16-1/4CA223J	RES CA 22K OHM 1/4W
R146	16-1/4CR220J	RES CR 22 OHM 1/4W
R201	16-1/4CR471J	RES CR 470 OHM 1/4W
R202	16-1/4CR101J	RES CR 100 OHM 1/4W
R204	16-1/4CA102J	RES CA 1K OHM 1/4W
R205	16-1/4CL104J	RES CL 100K OHM 1/4W
R206	16-1/4CA472J	RES CA 4K7 OHM 1/4W
R207	16-1/4CS271J	RES CS 270 OHM 1/4W
R208	16-1/4CA822J	RES CA 8K2 OHM 1/4W
R209	16-1/4CA103J	RES CA 10K OHM 1/4W
R210	16-1/4CA102J	RES CA 1K OHM 1/4W
R211	16-1/4CR104J	RES CR 100K OHM 1/4W
R212	16-1/4CR104J	RES CR 100K OHM 1/4W
R213	16-1/4CR563J	RES CR 56K OHM 1/4W
R214	16-1/4CR104J	RES CR 100K OHM 1/4W
R216	16-1/4CR223J	RES CR 22K OHM 1/4W
R217	16-1/4CA471J	RES CA 470 OHM 1/4W
R218	16-1/4CR103J	RES CR 10K OHM 1/4W
R219	16-1/4CR471J	RES CR 470 OHM 1/4W
R220	16-1/4CR152J	RES CR 1K5 OHM 1/4W
R221	16-1/4CA183J	RES CA 18K OHM 1/4W
R223	16-1/4CA103J	RES CA 10K OHM 1/4W

PEF. NO.	PART NO.	DESCRIPTION
RESISTORS (All are carbon film 1/4W +-5% unless noted otherwise)		
R224	16-1/4CA473J	RES CA 47K OHM 1/4W
R226	16-1/4CA103J	RES CA 10K OHM 1/4W
R227	16-1/4CA473J	RES CA 47K OHM 1/4W
R228	16-1/4CL102J	RES CL 1K OHM 1/4W
R229	16-1/4CU392J	RES CU 3K9 OHM 1/4W
R230	16-1/4CR474J	RES CR 470K OHM 1/4W
R237	16-1/4CR473J	RES CR 47K OHM 1/4W
R238	16-1/4CA102J	RES CA 1K OHM 1/4W
R239	16-1/4CA681J	RES CA 680 OHM 1/4W
R240	16-1/4CA681J	RES CA 680 OHM 1/4W
R241	16-1/4CA681J	RES CA 680 OHM 1/4W
R242	16-1/4CA681J	RES CA 680 OHM 1/4W
R243	16-1/4CA681J	RES CA 680 OHM 1/4W
R244	16-1/4CA270J	RES CA 27 OHM 1/4W
R245	16-1/4CA270J	RES CA 27 OHM 1/4W
R246	16-1/4CA270J	RES CA 27 OHM 1/4W
R247	16-1/4CA270J	RES CA 27 OHM 1/4W
R248	16-1/4CA270J	RES CA 27 OHM 1/4W
R249	16-1/4CA270J	RES CA 27 OHM 1/4W
R250	16-1/4CS101J	RES CS 27 OHM 1/4W
R251	16-1/4CL103J	RES CL 10K OHM 1/4W
R252	16-1/4CA122J	RES CA 1K2 OHM 1/4W
R253	16-1/4CA821J	RES CA 820 OHM 1/4W
R254	16-1/4CA122J	RES CA 1K2 OHM 1/4W
R255	16-1/4CA102J	RES CA 1K OHM 1/4W
R256	16-1/4CA103J	RES CA 10K OHM 1/4W
R257	16-1/4CA103J	RES CA 10K OHM 1/4W
R258	16-1/4CA103J	RES CA 10K OHM 1/4W
R259	16-1/4CA103J	RES CA 10K OHM 1/4W
R260	16-1/4CR473J	RES CR 47K OHM 1/4W
R261	16-1/4CS223J	RES CS 22K OHM 1/4W
R262	16-1/4CU183J	RES CU 18K OHM 1/4W
R263	16-1/4CA152J	RES CA 1K5 OHM 1/4W
R264	16-1/4CA102J	RES CA 1K OHM 1/4W
R265	16-1/4CR104J	RES CR 100K OHM 1/4W
R266	16-1/4CA470J	RES CA 47 OHM 1/4W
R301	16-1/4CR104J	RES CR 100K OHM 1/4W
R302	16-1/4CR104J	RES CR 100K OHM 1/4W
R303	16-1/4CR473J	RES CR 47K OHM 1/4W
R304	16-1/4CR473J	RES CR 47K OHM 1/4W
R305	16-1/4CA472J	RES CA 4K7 OHM 1/4W
R306	16-1/4CR472J	RES CR 4K7 OHM 1/4W
R307	16-1/4CR154J	RES CR 150K OHM 1/4W
R308	16-1/4CR154J	RES CR 150K OHM 1/4W
R309	16-1/4CA332J	RES CA 3K3 OHM 1/4W
R310	16-1/4CR332J	RES CR 3K3 OHM 1/4W
R311	16-1/4CR101J	RES CR 100 OHM 1/4W
R312	16-1/4CR101J	RES CR 100 OHM 1/4W
R313	16-1/4CR273J	RES CR 27K OHM 1/4W
R314	16-1/4CA273J	RES CA 27K OHM 1/4W
R315	16-1/4CR102J	RES CR 1K OHM 1/4W
R316	16-1/4CA102J	RES CA 1K OHM 1/4W
R317	16-1/4CA103J	RES CA 10K OHM 1/4W
R318	16-1/4CA103J	RES CA 10K OHM 1/4W
R319	16-1/4CR273J	RES CR 27K OHM 1/4W
R320	16-1/4CR273J	RES CR 27K OHM 1/4W
R321	16-1/4CR563J	RES CR 56K OHM 1/4W
R322	16-1/4CR563J	RES CR 56K OHM 1/4W
R323	16-1/4CA683J	RES CA 68K OHM 1/4W
R324	16-1/4CA683J	RES CA 68K OHM 1/4W
R325	16-1/4CA103J	RES CA 10K OHM 1/4W
R326	16-1/4CA103J	RES CA 10K OHM 1/4W

REF.NO.	PART NO.	DESCRIPTION
RESISTORS (All are carbon film 1/4W +-5% unless noted othise)		
R327	16-1/4CA102J	RES CA 1K OHM 1/4W
R328	16-1/4CA102J	RES CA 1K OHM 1/4W
R329	16-1/4CA102J	RES CA 1K OHM 1/4W
R330	16-1/4CA102J	RES CA 1K OHM 1/4W
R331	16-1/4CA153J	RES CA 15K OHM 1/4W
R332	16-1/4CA153J	RES CA 15K OHM 1/4W
R333	16-1/4CA102J	RES CA 1K OHM 1/4W
R334	16-1/4CA102J	RES CA 1K OHM 1/4W
R335	16-1/4CR103J	RES CR 10K OHM 1/4W
R336	16-1/4CR103J	RES CR 10K OHM 1/4W
R337	16-1/4CA103J	RES CA 10K OHM 1/4W
R338	16-1/4CA103J	RES CA 10K OHM 1/4W
R339	16-1/4CR103J	RES CR 10K OHM 1/4W
R340	16-1/4CA103J	RES CA 10K OHM 1/4W
R341	16-1/4CA223J	RES CA 22K OHM 1/4W
R342	16-1/4CA223J	RES CA 22K OHM 1/4W
R343	16-1/4CR223J	RES CR 22K OHM 1/4W
R344	16-1/4CR223J	RES CR 22K OHM 1/4W
R345	16-1/4CR681J	RES CR 680 OHM 1/4W
R346	16-1/4CR681J	RES CR 680 OHM 1/4W
R347	16-1/4CA473J	RES CA 47K OHM 1/4W
R348	16-1/4CR473J	RES CR 47K OHM 1/4W
R349	16-1/4CR473J	RES CR 47K OHM 1/4W
R350	16-1/4CR473J	RES CR 47K OHM 1/4W
R351	16-1/4CR104J	RES CR 100K OHM 1/4W
R352	16-1/4CR332J	RES CR 3K3 OHM 1/4W
R353	16-1/4CR332J	RES CR 3K3 OHM 1/4W
R354	16-1/4CR473J	RES CR 47K OHM 1/4W
R355	16-1/4CR473J	RES CR 47K OHM 1/4W
R356	16-1/4CA183J	RES CA 18K OHM 1/4W
R357	16-1/4CA103J	RES CA 10K OHM 1/4W
R358	16-1/4CR103J	RES CR 10K OHM 1/4W
R359	16-1/4CA221J	RES CA 220 OHM 1/4W
R360	16-1/4CA332J	RES CA 3K3 OHM 1/4W
R361	16-1/4CA221J	RES CA 220 OHM 1/4W
R362	16-1/4CA103J	RES CA 10K OHM 1/4W
R363	16-1/4CR223J	RES CR 22K OHM 1/4W
R364	16-1/4CA102J	RES CA 1K OHM 1/4W
R365	16-1/4CA103J	RES CA 10K OHM 1/4W
R366	16-1/4CA103J	RES CA 10K OHM 1/4W
R367	16-1/4CA102J	RES CA 1K OHM 1/4W
R368	16-1/4CA101J	RES CA 100 OHM 1/4W
R369	16-1/4CA101J	RES CA 100 OHM 1/4W
R401	16-1/4CR473J	RES CR 47K OHM 1/4W
R402	16-1/4CA392J	RES CA 3K9 OHM 1/4W
R404	16-1/4CR820J	RES CR 82 OHM 1/4W
R405	16-1/4CA220J	RES CA 22 OHM 1/4W
R406	16-1/4CA473J	RES CA 47K OHM 1/4W
R408	16-1/4CA472J	RES CA 4K7 OHM 1/4W
R409	16-1/4CR332J	RES CR 3K3 OHM 1/4W
R410	16-1/4CR471J	RES CR 470 OHM 1/4W
R411	16-1/4CA221J	RES CA 220 OHM 1/4W
R412	16-1/4CR152J	RES CR 1K5 OHM 1/4W
R413	16-1/4CR102J	RES CR 1K OHM 1/4W
R414	16-1/4CA564J	RES CA 560K OHM 1/4W
R415	16-1/4CR152J	RES CR 1K5 OHM 1/4W
R416	16-1/4CR221J	RES CR 220 OHM 1/4W
R417	16-1/4CR473J	RES CR 47K OHM 1/4W
R501	16-1/4CA101J	RES CA 100 OHM 1/4W
R502	16-1/4CA101J	RES CA 100 OHM 1/4W
R503	16-1/4CA103J	RES CA 10K OHM 1/4W
R504	16-1/4CA103J	RES CA 10K OHM 1/4W

REF. NO.	PART NO.	DESCRIPTION
RESISTORS (All are carbon film 1/4W +-5% unless noted otherwise)		
R505	16-1/4CA104J	RES CA 100K OHM 1/4W
R506	16-1/4CA104J	RES CA 100K OHM 1/4W
R507	16-1/4CA104J	RES CA 100K OHM 1/4W
R508	16-1/4CA104J	RES CA 100K OHM 1/4W
R509	16-1/4CA562J	RES CA 5K6 OHM 1/4W
R510	16-1/4CA562J	RES CA 5K6 OHM 1/4W
R511	16-1/4CA562J	RES CA 5K6 OHM 1/4W
R512	16-1/4CA562J	RES CA 5K6 OHM 1/4W
R513	16-1/4CR820J	RES CR 82 OHM 1/4W
R514	16-1/4CR820J	RES CR 82 OHM 1/4W
R515	16-1/4CR393J	RES CR 39K OHM 1/4W
R516	16-1/4CR393J	RES CR 39K OHM 1/4W
R517	16-1/4CA681J	RES CA 680 OHM 1/4W
R518	16-1/4CR681J	RES CR 680 OHM 1/4W
R521	16-1/4CA332J	RES CA 3K3 OHM 1/4W
R522	16-1/4CA332J	RES CA 3K3 OHM 1/4W
R523	16-1/4CA392J	RES CA 3K9 OHM 1/4W
R525	16-1/4CA154J	RES CA 150K OHM 1/4W
R526	16-1/4CA154J	RES CA 150K OHM 1/4W
R527	16-1/4CA472J	RES CA 4K7 OHM 1/4W
R528	16-1/4CA472J	RES CA 4K7 OHM 1/4W
R529	16-1/4CR104J	RES CR 100K OHM 1/4W
R530	16-1/4CR104J	RES CR 100K OHM 1/4W
R531	16-1/4CA562J	RES CA 5K6 OHM 1/4W
R532	16-1/4CA562J	RES CA 5K6 OHM 1/4W
R533	16-1/4CA562J	RES CA 5K6 OHM 1/4W
R534	16-1/4CA562J	RES CA 5K6 OHM 1/4W
R537	16-1/4CA103J	RES CA 10K OHM 1/4W
R538	16-1/4CA103J	RES CA 10K OHM 1/4W
R539	16-1/4CA221J	RES CA 220 OHM 1/4W
R540	16-1/4CA221J	RES CA 220 OHM 1/4W
R541	16-1/4CA103J	RES CA 10K OHM 1/4W
R542	16-1/4CA103J	RES CA 10K OHM 1/4W
R543	16-1/4CA124J	RES CA 120K OHM 1/4W
R544	16-1/4CA124J	RES CA 120K OHM 1/4W
R545	16-1/4CA181J	RES CA 180 OHM 1/4W
R546	16-1/4CA181J	RES CA 180 OHM 1/4W
R547	16-1/4CA473J	RES CA 47K OHM 1/4W
R548	16-1/4CA473J	RES CA 47K OHM 1/4W
R549	16-1/4CA681J	RES CA 680 OHM 1/4W
R550	16-1/4CA681J	RES CA 680 OHM 1/4W
R551	16-1/4CA102J	RES CA 1K OHM 1/4W
R552	16-1/4CA102J	RES CA 1K OHM 1/4W
R553	16-1/4CA102J	RES CA 1K OHM 1/4W
R554	16-1/4CA102J	RES CA 1K OHM 1/4W
R601	16-1/4CA183J	RES CA 18K OHM 1/4W
R602	16-1/4CA183J	RES CA 18K OHM 1/4W
R603	16-1/4CA473J	RES CA 47K OHM 1/4W
R604	16-1/4CA473J	RES CA 47K OHM 1/4W
R605	16-1/4CA681J	RES CA 680 OHM 1/4W
R606	16-1/4CA681J	RES CA 680 OHM 1/4W
R607	16-1/4CA152J	RES CA 1K5 OHM 1/4W
R608	16-1/4CA152J	RES CA 1K5 OHM 1/4W
R609	16-1/4CA152J	RES CA 1K5 OHM 1/4W
R610	16-1/4CA152J	RES CA 1K5 OHM 1/4W
R611	16-1/4CA391J	RES CA 390 OHM 1/4W
R612	16-1/4CA391J	RES CA 390 OHM 1/4W
R613	16-1/4CA562J	RES CA 5K6 OHM 1/4W
R614	16-1/4CA562J	RES CA 5K6 OHM 1/4W
R615	16-1/4CA392J	RES CA 3K9 OHM 1/4W
R616	16-1/4CA392J	RES CA 3K9 OHM 1/4W
R617	16-1/4CA272J	RES CA 2K7 OHM 1/4W

REF. NO.	PART NO.	DESCRIPTION
RESISTORS (All are carbon film 1/4W +-5% unless noted otherwise)		
R618	16-1/4CA272J	RES CA 2K7 OHM 1/4W
R619	16-1AS682J	RES MO SWL 6K8 OHM 1W
R620	16-1AS682J	RES MO SWL 6K8 OHM 1W
R621	16-1/4CA151J	RES CA 150 OHM 1/4W
R622	16-1/4CA151J	RES CA 150 OHM 1/4W
R623	16-1/4CA182J	RES CA 1K8 OHM 1/4W
R624	16-1/4CA182J	RES CA 1K8 OHM 1/4W
R625	16-1/4CA681J	RES CA 680 OHM 1/4W
R626	16-1/4CA681J	RES CA 680 OHM 1/4W
R627	16-1/4CA680J	RES CA 68 OHM 1/4W
R628	16-1/4CA680J	RES CA 68 OHM 1/4W
R629	16-1/4CA680J	RES CA 68 OHM 1/4W
R630	16-1/4CA680J	RES CA 68 OHM 1/4W
R631	16-1/4AS151J	RES MO SWL 150 OHM 1/4W
R632	16-1/4AS151J	RES MO SWL 150 OHM 1/4W
R633	16-1/4CA473J	RES CA 47K OHM 1/4W
R634	16-1/4CA473J	RES CA 47K OHM 1/4W
R635	16-1042	RES CM 0.1 OHM 3W
R636	16-1042	RES CM 0.1 OHM 3W
R637	16-1AS100J	RES MO SWL 10 OHM 1W
R638	16-1AS100J	RES MO SWL 10 OHM 1W
R639	16-1AS3R3J	RES MO SWL 3.3 OHM 1W
R640	16-1AS3R3J	RES MO SWL 3.3 OHM 1W
R641	16-1AS471J	RES MO SWL 470 OHM 1W
R642	16-1AS471J	RES MO SWL 470 OHM 1W
R643	16-1/4CA223J	RES CA 22K OHM 1/4W
R651	16-1/4CA681J	RES CA 680 OHM 1/4W
R652	16-1/4CA681J	RES CA 680 OHM 1/4W
R653	16-1/4CA823J	RES CA 82K OHM 1/4W
R654	16-1/4CA823J	RES CA 82K OHM 1/4W
R655	16-1/4CA823J	RES CA 82K OHM 1/4W
R656	16-1/4CA823J	RES CA 82K OHM 1/4W
R657	16-1/4CA682J	RES CA 6K8 OHM 1/4W
R658	16-1/4CA682J	RES CA 6K8 OHM 1/4W
R661	16-1/4CR153J	RES CR 15K OHM 1/4W
R662	16-1/4CR153J	RES CR 15K OHM 1/4W
R670	16-1/4CA272J	RES CA 2K7 OHM 1/4W
R671	16-1/4CA563J	RES CA 56K OHM 1/4W
R701	16-1/4CA823J	RES CA 82K OHM 1/4W
R702	16-1/4CA823J	RES CA 82K OHM 1/4W
R703	16-1/4CA223J	RES CA 22K OHM 1/4W
R704	16-1/4CA223J	RES CA 22K OHM 1/4W
R751	16-1/4CA152J	RES CA 1K5 OHM 1/4W
R752	16-1/4CA152J	RES CA 1K5 OHM 1/4W
R753	16-1/4CA153J	RES CA 15K OHM 1/4W
R754	16-1/4CA153J	RES CA 15K OHM 1/4W
R755	16-1/4CA563J	RES CA 56K OHM 1/4W
R756	16-1/4CA563J	RES CA 56K OHM 1/4W
R757	16-1/4CA562J	RES CA 5K6 OHM 1/4W
R758	16-1/4CA562J	RES CA 5K6 OHM 1/4W
R759	16-1/4CA682J	RES CA 6K8 OHM 1/4W
R760	16-1/4CA103J	RES CA 10K OHM 1/4W
R761	16-1/4CA472J	RES CA 4K7 OHM 1/4W
R762	16-1/4CA682J	RES CA 6K8 OHM 1/4W
R763	16-1/4CA224J	RES CA 220K OHM 1/4W
R764	16-1/4CA101J	RES CR 100 OHM 1/4W
R765	16-1/4CA103J	RES CA 10K OHM 1/4W
R803	16-1/4CA681J	RES CA 680 OHM 1/4W
R804	16-1/4CA681J	RES CA 680 OHM 1/4W
R805	16-1/4CA473J	RES CA 47K OHM 1/4W
R806	16-1/4CA473J	RES CA 47K OHM 1/4W
R809	16-1/4CA272J	RES CA 2K7 OHM 1/4W
R810	16-1/4CA272J	RES CA 2K7 OHM 1/4W

REF. NO.	PART NO.	DESCRIPTION
RESISTORS (All are carbon film 1/4W +-5% unless noted otherwise)		
R811	16-1/4CA153J	RES CA 15K OHM 1/4W
R812	16-1/4CA153J	RES CA 15K OHM 1/4W
R813	16-1/4CA472J	RES CA 4K7 OHM 1/4W
R814	16-1/4CA472J	RES CA 4K7 OHM 1/4W
R815	16-1/4CA472J	RES CA 4K7 OHM 1/4W
R816	16-1/4CA472J	RES CA 4K7 OHM 1/4W
R817	16-1/4CA222J	RES CA 2K2 OHM 1/4W
R818	16-1/4CA222J	RES CA 2K2 OHM 1/4W
R819	16-1/4CA152J	RES CA 1K5 OHM 1/4W
R820	16-1/4CA152J	RES CA 1K5 OHM 1/4W
R821	16-1/4CA152J	RES CA 1K5 OHM 1/4W
R822	16-1/4CA152J	RES CA 1K5 OHM 1/4W
R823	16-1/4CA102J	RES CA 1K OHM 1/4W
R824	16-1/4CA102J	RES CA 1K OHM 1/4W
R825	16-1/4CA152J	RES CA 1K5 OHM 1/4W
R826	16-1/4CA152J	RES CA 1K5 OHM 1/4W
R827	16-1/4CA153J	RES CA 15K OHM 1/4W
R828	16-1/4CA153J	RES CA 15K OHM 1/4W
R829	16-1/4CA473J	RES CA 47K OHM 1/4W
R830	16-1/4CA473J	RES CA 47K OHM 1/4W
R831	16-1/4CA272J	RES CA 2K7 OHM 1/4W
R832	16-1/4CA272J	RES CA 2K7 OHM 1/4W
R833	16-1/4CA682J	RES CA 6K8 OHM 1/4W
R834	16-1/4CA682J	RES CA 6K8 OHM 1/4W
R835	16-1/4CA103J	RES CA 10K OHM 1/4W
R836	16-1/4CA103J	RES CA 10K OHM 1/4W
R838	16-1/4CA821J	RES CA 820 OHM 1/4W
R839	16-1/4CA682J	RES CA 6K8 OHM 1/4W
R840	16-1/4CA682J	RES CA 6K8 OHM 1/4W
R843	16-1/4CA681J	RES CA 680 OHM 1/4W
R844	16-1/4CA681J	RES CA 680 OHM 1/4W
R845	16-1/4CA681J	RES CA 680 OHM 1/4W
R846	16-1/4CA681J	RES CA 680 OHM 1/4W
R847	16-1/4CA223J	RES CA 22K OHM 1/4W
R848	16-1/4CA223J	RES CA 22K OHM 1/4W
R849	16-1/4CR104J	RES CR 100K OHM 1/4W
R850	16-1/4CR104J	RES CR 100K OHM 1/4W
R901	16-1/4CA222J	RES CA 2K2 OHM 1/4W
R902	16-1/4CA222J	RES CA 2K2 OHM 1/4W
R903	16-1/4CA103J	RES CA 10K OHM 1/4W
R904	16-1/4CA103J	RES CA 10K OHM 1/4W
R905	16-1/4CA682J	RES CA 6K8 OHM 1/4W
R906	16-1/4CA682J	RES CA 6K8 OHM 1/4W
R907	16-1/4CA222J	RES CA 2K2 OHM 1/4W
R908	16-1/4CA222J	RES CA 2K2 OHM 1/4W
R909	16-1/4CA103J	RES CA 10K OHM 1/4W
R910	16-1/4CA103J	RES CA 10K OHM 1/4W
R911	16-1/4CA223J	RES CA 22K OHM 1/4W
R912	16-1/4CA223J	RES CA 22K OHM 1/4W
R913	16-1/4CA152J	RES CA 1K5 OHM 1/4W
R914	16-1/4CA152J	RES CA 1K5 OHM 1/4W
R915	16-1/4CA152J	RES CA 1K5 OHM 1/4W
R916	16-1/4CA152J	RES CA 1K5 OHM 1/4W
R921	16-1/4CA1R0J	RES CA 1 OHM 1/4W
R922	16-1/4CA1R0J	RES CA 1 OHM 1/4W
R923	16-1/4CA1R0J	RES CA 1 OHM 1/4W
R924	16-1/4CA1R0J	RES CA 1 OHM 1/4W
R925	16-1/4CA332J	RES CA 3K3 OHM 1/4W
R926	16-1/4CA103J	RES CA 10K OHM 1/4W
R927	16-1/4CA103J	RES CA 10K OHM 1/4W
R928	16-1/4CA103J	RES CA 10K OHM 1/4W
R929	16-1/4CA103J	RES CA 10K OHM 1/4W

REF. NO.	PART NO.	DESCRIPTION
CAPACITORS		
C001	17-5DR2R0DPH	CAP CER R 2pF 50V +-0.5pFPH
C002	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C003	17-5DR473Z	CAP CER R .047uF 50V +80-20%
C004	17-5DR473Z	CAP CER R .047uF 50V +80-20%
C005	17-5DR2R0DPH	CAP CER R 2pF 50V +-0.5pFPH
C006	17-5DR121K	CAP CER R 120pF 50V +-10%
C007	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C008	17-5DR2R0DPH	CAP CER R 2pH 50V +-0.5pFPH
C009	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C010	17-5DR101K	CAP CER R 100pF 50V +-10%
C011	17-5DR181KCH	CAP CER R 180pF 50V +-10% CH
C012	17-5DR473Z	CAP CER R .047uF 50V +80-20%
C013	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C014	17-5DR470KCH	CAP CER R 47pF 50V +-10% CH
C015	17-5DR2R0DPH	CAP CER R 2pF 50V +-0.5pFPH
C016	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C017	17-5DR100DCH	CAP CER R 10pF 50V +-0.5pFCH
C018	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C019	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C020	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C021	17-5DR220KCH	CAP CER R 22pF 50V +-10% CH
C101	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C102	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C103	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C104	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C105	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C106	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C107	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C108	17-5DF223Z	CAP CER SM .022uF 50V +80-20%
C109	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C110	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C111	17-5DR473Z	CAP CER R .047uF 50V +80-20%
C112	17-5DR473Z	CAP CER R .047uF 50V +80-20%
C113	17-1.6EF227Y	CAP EL SM 220uF 16V +50-10%
C114	17-5ER105Y	CAP EL SM 1uF 50V +50-10%
C115	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C116	17-5ER105Y	CAP EL R 1uF 50V +50-10%
C117	17-1.6EF107Y	CAP EL SM 100uF 16V +50-10%
C118	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C119	17-5ER474Y	CAP EL R .47uF 50V +50-10%
C120	17-5FR103J	CAP MY R .01uF 50V +-5%
C121	17-5DR221K	CAP CER R 220pF 50V +-10%
C122	17-5SR335M-1	CAP ELBP 3.3uF 50V +-20%
C123	17-5SR105M-1	CAP ELBP 1uF 50V +-20%
C124	17-5ER475Y	CAP EL R 4.7uF 50V +50-10%
C125	17-1.6EF227Y	CAP EL SM 220uF 16V +50-10%
C126	17-2.5ER106Y	CAP EL R 10uF 25V +50-10%
C127	17-5FR473J	CAP MY R .047uF 50V +-5%
C128	17-5DR102Z	CAP CER R .001uF 50V +80-20%
C129	17-1.6OF335M	CAP TAS 3.3uF 16V +-20%
C130	17-5DR431J	CAP CER R 430pF 50V +-5%
C131	17-5DR431J	CAP CER R 430pF 50V +-5%
C132	17-5DR201K	CAP CER R 200pF 50V +-10%
C133	17-5DR201K	CAP CER R 200pF 50V +-10%
C134	17-2.5ER106Y	CAP EL R 10uF 25V +50-10%
C135	17-2.5ER106Y	CAP EL R 10uF 25V +50-10%
C136	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C137	17-2.5ER106Y	CAP EL R 10uF 25V +50-10%
C138	17-2.5EF106Y	CAP EL SM 10uF 25V +50-10%
C139	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C140	17-5DR101K	CAP CER R 100pF 50V +-10%
C142	17-1.6ER107Y	CAP EL R 100uF 16V +50-10%

REF. NO.	PART NO.	DESCRIPTION
CAPACITORS		
C144	17-5DR473Z	CAP CER R .047uF 50V +80-20%
C145	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C201	17-5DR103M	CAP CER R .01uF 50V +-20%
C202	17-5DR103M	CAP CER R .01uF 50V +-20%
C203	17-0.63E107Y	CAP EL SM 100uF 6.3V +50-10%
C204	17-5DR473Z	CAP CER R .047uF 50V +80-20%
C205	17-5DR103M	CAP CER R .01uF 50V +-20%
C206	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C207	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C208	17-5DF223Z	CAP CER SM .022uF 50V +80-20%
C209	17-1.6OF224M	CAP TAS .22uF 16V +-20%
C210	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C211	17-0.63EF478Y	CAP EL SM 4700uF 6.3V +50-10%
C212	17-5DR470KCH	CAP CER R 47pF 50V +-10% CH
C213	17-5DR150KSH	CAP CER R 15pF 50V +-10% SH
C214	17-5EF105Y	CAP EL SM 1uF 50V +50-10%
C215	17-5ER105Y	CAP EL R 1uF 50V +50-10%
C216	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C217	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C218	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C219	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C220	17-5ER106Y	CAP EL R 10uF 50V +50-10%
C221	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C222	17-5ER106Y	CAP EL R 10uF 50V +50-10%
C223	17-5FF473J	CAP MY SM .047uF 50V +-5%
C224	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C225	17-0.63E107Y	CAP EL SM 100uF 6.3V +50-10%
C226	17-1.6EG228Y	CAP EL EG 2200uF 16V +50-10%
C227	17-1.6ER107Y	CAP EL R 100uF 16V +50-10%
C228	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C229	17-0.63EF228Y	CAP EL SM 2200uF 6.3V +50-10%
C230	17-1.6ER336Y	CAP EL SM 33uF 16V +50-10%
C231	17-2.5ER106Y	CAP EL SM 10uF 25V +50-10%
C232	17-1.6ER336Y	CAP EL SM 33uF 16V +50-10%
C233	17-2.5ER106Y	CAP EL SM 10uF 25V +50-10%
C234	17-5DR103M	CAP CER R .01uF 50V +-20%
C235	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C236	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C237	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C238	17-5ER476Y	CAP EL R 47uF 50V +50-10%
C301	17-2.5ER106Y	CAP EL R 10uF 25V +50-10%
C302	17-2.5ER106Y	CAP EL R 10uF 25V +50-10%
C303	17-5FR153J	CAP MY R .015uF 50V +-5%
C304	17-5FR153J	CAP MY R .015uF 50V +-5%
C305	17-2.5ER106Y	CAP EL R 10uF 25V +50-10%
C306	17-2.5ER106Y	CAP EL R 10uF 25V +50-10%
C307	17-5FR223Z	CAP MY R .022uF 50V +-5%
C308	17-5FR223Z	CAP MY R .022uF 50V +-5%
C309	17-5DR221K	CAP CER R 220pF 50V +-10%
C310	17-5DR221K	CAP CER R 220pF 50V +-10%
C311	17-2.5ER106Y	CAP EL R 10uF 25V +50-10%
C312	17-2.5ER106Y	CAP EL R 10uF 25V +50-10%
C313	17-2.5ER106Y	CAP EL R 10uF 25V +50-10%
C314	17-2.5ER106Y	CAP EL R 10uF 25V +50-10%
C315	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C316	17-5ER474Y	CAP EL R .47uF 50V +50-10%
C317	17-0.63ER477Y	CAP EL R 220uF 6.3V +50-10%
C318	17-1.6ER476Y	CAP EL R 47uF 16V +50-10%
C319	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C320	17-5ER105Y	CAP EL R 1uF 50V +50-10%
C321	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C322	17-1.6ER476Y	CAP CER R 47uF 16V +50-10%

REF. NO.	PART NO.	DESCRIPTION
CAPACITORS		
C323	17-5DR220KCH	CAP CER R 22pF 50V +50-10% CH
C324	17-5DR220KCH	CAP CER R 22pF 50V +50-10% CH
C325	17-5DR220KCH	CAP CER R 22pF 50V +50-10% CH
C326	17-5DR220KCH	CAP CER R 22pF 50V +50-10% CH
C401	17-5DR473Z	CAP CER R .047uF 50V +80-20%
C402	17-5DR150KCH	CAP CER R 15pF 50V +-10% CH
C403	17-5U391J	CAP PS 390pF 50V +-5%
C404	17-5DR473Z	CAP CER R .047uF 50V +80-20%
C405	17-1.6ER226Y	CAP EL R 22uF 16V +50-10%
C406	17-5ER474Y	CAP EL R .47uF 50V +80-20%
C407	17-5DR473Z	CAP CER R .047uF 50V +80-20%
C408	17-5FR103J	CAP MY R .01uF 50V +-5%
C409	17-5FF224J	CAP MY F .22uF 50V +-5%
C410	17-5FR103J	CAP MY R .01uF 50V +-5%
C411	17-5DR102M	CAP CER R .001uF 50V +-20%
C412	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C413	17-1.6ER107Y	CAP EL R 100uF 16V +50-10%
C414	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C415	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C416	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C417	17-5DR473Z	CAP CER R .047uF 50V +80-20%
C418	17-5DR102M	CAP CER R .001uF 50V +80-20%
C419	17-5DR100KCH	CAP CER R 10pF 50V +-10% CH
C420	17-5DR223Z	CAP CER R .022uF 50V +80-20%
C421	17-1.6EF477Y	CAP EL EF 470uF 16V +50-10%
C422	17-5DR103M	CAP CER R .01uF 50V +80-20%
C501	17-5DR102M	CAP CER R 1000pF 50V +-20%
C502	17-5DR102M	CAP CER R 1000pF 50V +-20%
C503	17-1.6ER476Y	CAP EL R 47uF 16V +50-10%
C504	17-1.6ER476Y	CAP EL R 47uF 16V +50-10%
C505	17-1.6ER476Y	CAP EL R 47uF 16V +50-10%
C506	17-1.6ER476Y	CAP EL R 47uF 16V +50-10%
C507	17-2062	CAP EL R 4.7uF 16V +50-10%
C508	17-2062	CAP EL R 4.7uF 16V +50-10%
C509	17-2062	CAP EL R 4.7uF 16V +50-10%
C510	17-2062	CAP EL R 4.7uF 16V +50-10%
C511	17-5DR561M6Y	CAP CER R 560pF 50V +-20%
C512	17-5DR561M6Y	CAP CER R 560pF 50V +-20%
C515	17-1.6ER227Y	CAP EL R 220uF 16V +50-10%
C516	17-1.6ER227Y	CAP EL R 220uF 16V +50-10%
C517	17-2062	CAP EL R 4.7uF 16V +50-10%
C518	17-2062	CAP EL R 4.7uF 16V +50-10%
C519	17-5DR101M	CAP CER R 100pF 50V +-20%
C520	17-5DR101M	CAP CER R 100PF 50V +-20%
C521	17-5DR221M	CAP CER R 220pF 50V +-20%
C522	17-5DR221M	CAP CER R 220pF 50V +-20%
C523	17-5DR101M	CAP CER R 100PF 50V +-20%
C524	17-5DR101M	CAP CER R 100PF 50V +-20%
C527	17-5DR470M	CAP CER R 47pF 50V +-20%
C528	17-5DR470M	CAP CER R 47pF 50V +-20%
C529	17-6.3ER227Y	CAP EL R 220uF 6.3V +50-10%
C530	17-6.3ER227Y	CAP EL R 220uF 6.3V +50-10%
C531	17-5FR822J	CAP MY R.0082uF 50V +-5%
C532	17-5FR822J	CAP MY R.0082uF 50V +-5%
C533	17-5FR273J	CAP MY R .027uF 50V +-5%
C534	17-5FR273J	CAP MY R .027uF 50V +-5%
C535	17-2.5ER226Y	CAP EL R 22uF 25V +50-10%
C536	17-2.5ER226Y	CAP EL R 22uF 25V +50-10%
C537	17-5FR152J	CAP MY R.0015uF 50V +-5%
C538	17-5FR152J	CAP MY R.0015uF 50V +-5%
C539	17-5MR104J	CAP MMY R .1uF 50V +-5%
C601	17-5HR474J	CAP MMY R .47uF 50V +-5%

REF. NO.	PART NO.	DESCRIPTION
CAPACITORS		
C602	17-5HR474J	CAP MMY R .47uF 50V +-5%
C603	17-5HR474J	CAP MMY R .47uF 50V +-5%
C604	17-5HR474J	CAP MMY R .47uF 50V +-5%
C605	17-5DR221M	CAP CER R 220pF 50V +-20%
C606	17-5DR221M	CAP CER R 220pF 50V +-20%
C607	17-1ER476Y	CAP EL R 47uF 10V +50-10%
C608	17-1ER476Y	CAP EL R 47uF 10V +50-10%
C609	17-5DR101M	CAP CER R 100pF 50V +-20%
C610	17-5DR101M	CAP CER R 100pF 50V +-20%
C611	17-5DR101M	CAP CER R 100pF 50V +-20%
C612	17-5DR101M	CAP CER R 100pF 50V +-20%
C613	17-5DR221M	CA[CER R 220pF 50V +-20%
C614	17-5DR221M	CA[CER R 220pF 50V +-20%
C615	17-5ER105Y	CAP EL R 1uF 50V +50-10%
C616	17-5ER105Y	CAP EL R 1uF 50V +50-10%
C617	17-5DR100K	CAP CER R 10pF 50V +-10%
C618	17-5DR100K	CAP CER R 10pF 50V +-10%
C619	17-5ER105Y	CAP EL R 1uF 50V +50-10%
C620	17-5ER105Y	CAP EL R 1uF 50V +50-10%
C621	17-5ER105Y	CAP EL R 1uF 50V +50-10%
C622	17-5ER105Y	CAP EL R 1uF 50V +50-10%
C623	17-5HR104J	CAP MMY R .1uF 50V +-5%
C624	17-5HR104J	CAP MMY R .1uF 50V +-5%
C625	17-1ER476Y	CAP EL R 47uF 10V +50-10%
C627	17-5FR472J	CAP MY R .0047uF 50V +-5%
C628	17-5FR472J	CAP MY R .0047uF 50V +-5%
C651	17-2.5ER107Y	CAP EL R 100uF 25V +50-10%
C652	17-2.5ER107Y	CAP EL R 100uF 25V +50-10%
C653	17-2.5ER107Y	CAP EL R 100uF 25V +50-10%
C654	17-2.5ER107Y	CAP EL R 100uF 25V +50-10%
C655	17-5DR102M	CAP CER R 1000pF 50V +-20%
C656	17-5DR102M	CAP CER R 1000pF 50V +-20%
C657	17-5DR102M	CAP CER R 1000pF 50V +-20%
C658	17-5DR102M	CAP CER R 1000pF 50V +-20%
C661	17-5FR473J	CAP MY R .047uF 50V +-5%
C662	17-5FR473J	CAP MY R .047uF 50V +-5%
C701	17-1ER476Y	CAP EL R 47uF 10V +50-10%
C702	17-1ER476Y	CAP EL R 47uF 10V +50-10%
C703	17-1ER476Y	CAP EL R 47uF 10V +50-10%
C704	17-1ER476Y	CAP EL R 47uF 10V +50-10%
C751	17-5FR223J	CAP MY R .022uF 50V +-5%
C752	17-5FR223J	CAP MY R .022uF 50V +-5%
C753	17-2.5ER106Y	CAP EL R 10uF 25V +50-10%
C754	17-1ER476Y	CAP EL R 47uF 10V +50-10%
C755	17-1ER336Y	CAP EL R 33uF 10V +50-10%
C801	17-2.5ER106Y	CAP EL R 10uF 25V +50-10%
C802	17-2.5ER106Y	CAP EL R 10uF 25V +50-10%
C807	17-5DR221M	CAP CER R 220pF 50V +-20%
C808	17-5DR221M	CAP CER R 220pF 50V +-20%
C809	17-5DR101M	CAP CER R 100pF 50V +-20%
C810	17-5DR101M	CAP CER R 100pF 50V +-20%
C811	17-1ER476Y	CAP EL R 47uF 10V +50-10%
C812	17-1ER476Y	CAP EL R 47uF 10V +50-10%
C813	17-5HR104J	CAP MMY R .1uF 50V +-5%
C814	17-5HR104J	CAP MMY R .1uF 50V +-5%
C815	17-5HR104J	CAP MMY R .1uF 50V +-5%
C816	17-5HR104J	CAP MMY R .1uF 50V +-5%
C817	17-5FR682J	CAP MY R .0068uF 50V +-5%
C818	17-5FR682J	CAP MY R .0068uF 50V +-5%
C819	17-5DR101M	CAP CER R 100pF 50V +-20%
C820	17-5DR101M	CAP CER R 100pF 50V +-20%
C821	17-5HR474J	CAP MMY R .47uF 50V +-5%

REF. NO.	PART NO.	DESCRIPTION
CAPACITORS		
C822	17-5HR474J	CAP MMY R .47uF 50V +-5%
C823	17-5HR474J	CAP MMY R .47uF 50V +-5%
C824	17-5HR474J	CAP MMY R .47uF 50V +-5%
C825	17-5DR221M	CAP CER R 220pF 50V +-20%
C826	17-5DR221M	CAP CER R 220pF 50V +-20%
C827	17-5DR101M	CAP CER R 100pF 50V +-20%
C828	17-5DR101M	CAP CER R 100pF 50V +-20%
C829	17-1ER476Y	CAP EL R 47uF 10V +50-10%
C830	17-1ER476Y	CAP EL R 47uF 10V +50-10%
C831	17-5FR222J	CAP MY R .0022uF 50V +-5%
C832	17-5FR222J	CAP MY R .0022uF 50V +-5%
C833	17-5DR101M	CAP CER R 100pF 50V +-20%
C834	17-5DR101M	CAP CER R 100pF 50V +-20%
C835	17-5FR683J	CAP MY R .068uF 50V +-5%
C836	17-5FR683J	CAP MY R .068uF 50V +-5%
C837	17-1ER476Y	CAP EL R 47uF 10V +50-10%
C838	17-1ER476Y	CAP EL R 47uF 10V +50-10%
C839	17-5DR102M	CAP CER R 1000pF 50V +-20%
C840	17-5DR102M	CAP CER R 1000pF 50V +-20%
C841	17-1ER105Y	CAP EL R 1uF 50V +50-10%
C842	17-1ER105Y	CAP EL R 1uF 50V +50-10%
C901	17-8ER109Y	CAP EL R 10000uF 80V +50-10%
C902	17-8ER109Y	CAP EL R 10000uF 80V +50-10%
C903	17-5ER478Y	CAP EL R 4700uF 50V +50-10%
C904	17-5ER478Y	CAP EL R 4700uF 50V +50-10%
C905	17-5ER478Y	CAP EL R 4700uF 50V +50-10%
C906	17-5ER478Y	CAP EL R 4700uF 50V +50-10%
C907	17-8ER477Y	CAP EL R 470uF 80V +50-10%
C908	17-8ER477Y	CAP EL R 470uF 80V +50-10%
C909	17-2.5ER226Y	CAP EL R 22uF 25V +50-10%
C910	17-2.5ER226Y	CAP EL R 22uF 25V +50-10%
C911	17-2.5ER226Y	CAP EL R 22uF 25V +50-10%
C912	17-2.5ER226Y	CAP EL R 22uF 25V +50-10%
C913	17-6.3ER106Y	CAP EL R 10uF 63V +50-10%
C914	17-6.3ER106Y	CAP EL R 10uF 63V +50-10%
C915	17-3.5ER477Y	CAP EL R 470uF 35V +50-10%
C916	17-3.5ER477Y	CAP EL R 470uF 35V +50-10%
C917	17-2.5ER226Y	CAP EL R 22uF 25V +50-10%
C918	17-2.5ER226Y	CAP EL R 22uF 25V +50-10%
C919	17-2.5ER226Y	CAP EL R 22uF 25V +50-10%
C920	17-2.5ER226Y	CAP EL R 22uF 25V +50-10%
C921	17-2.5ER476Y	CAP EL R 47uF 25V +50-10%
C922	17-2.5ER476Y	CAP EL R 47uF 25V +50-10%
C923	17-2.5ER108Y	CAP EL R 1000uF 25V +50-10%
C924	17-2.5ER476Y	CAP EL R 47uF 25V +50-10%
C925	17-2.5ER476Y	CAP EL R 47uF 25V +50-10%
C927	17-2.5ER476Y	CAP EL R 47uF 25V +50-10%
C931	17-5FR103J	CAP MY R .01uF 50V +-5%
C932	17-5FR103J	CAP MY R .01uF 50V +-5%
C933	17-5FR103J	CAP MY R .01uF 50V +-5%
C934	17-5FR103J	CAP MY R .01uF 50V +-5%
C935	17-5FR103J	CAP MY R .01uF 50V +-5%
C936	17-5FR103J	CAP MY R .01uF 50V +-5%
C937	17-5FR103J	CAP MY R .01uF 50V +-5%
C938	17-5FR103J	CAP MY R .01uF 50V +-5%
C939	17-5FR103J	CAP MY R .01uF 50V +-5%
C940	17-5FR103J	CAP MY R .01uF 50V +-5%
C941	17-5FR103J	CAP MY R .01uF 50V +-5%
C942	17-5FR103J	CAP MY R .01uF 50V +-5%
C943	17-5FR103J	CAP MY R .01uF 50V +-5%
C944	17-5FR103J	CAP MY R .01uF 50V +-5%
C945	17-5FR103J	CAP MY R .01uF 50V +-5%

REF. NO.	PART NO.	DESCRIPTION
CAPACITORS		
C946	17-5FR103J	CAP MY R .01uF 50V +-5%
C947	17-2.5ER226Y	CAP EL R 22uF 25V +50-10%
C948	17-2.5ER226Y	CAP EL R 22uF 25V +50-10%
VARIABLE RESISTORS		
VR101	VR01-F08103B	SEMI-FIX 8ø 10KB FLAT
VR102	VR01-F08203B	SEMI-FIX 8ø 20KB FLAT
VR103	VR03-F08104B	SEMI-FIX 8ø 100KB FLAT
VR301	VR01-F08104B	SEMI-FIX 8ø 100KB FLAT
VR302	VR01-F08503B	SEMI-FIX 8ø 50KB FLAT
VR303	VR01-F08203B	SEMI-FIX 8ø 20KB FLAT
VR401	VR01-F08203B	SEMI-FIX 8ø 20KB FLAT
VR601	VR03-F08501B	SEMI-FIX 8ø 500 OHM B FLAT
VR602	VR03-F08501B	SEMI-FIX 8ø 500 OHM B FLAT
VR801	VR03-F08202B	SEMI-FIX 8ø 500 OHM 2KB FLAT
VARIABLE CAPACITORS		
VD001	30-1082*A	VARI CAP BB204B
VD002	30-1082*A	VARI CAP BB204B
VD003	30-1082*A	VARI CAP BB204B
VD004	30-1082*A	VARI CAP BB204B
VD401	30-1146	VARI CAP SVC333B
VD402	30-1146	VARI CAP SVC333B
DIODES		
D101	30-1152TA	1N4148 A-TYPE
D102	30-1152TA	1N4148 A-TYPE
D103	30-1152TA	1N4148 A-TYPE
D104	30-1152TA	1N4148 A-TYPE
D105	30-1010	1N60 A-TYPE
D106	30-1152TA	1N4148 A-TYPE
D107	30-1152TA	1N4148 A-TYPE
D108	30-1152TA	1N4148 A-TYPE
D109	30-1152TA	1N4148 A-TYPE
D110	30-1152TA	1N4148 A-TYPE
D111	30-1152TA	1N4148 A-TYPE
D201	30-1152TA	1N4148 A-TYPE
D202	30-1152TA	1N4148 A-TYPE
D206	30-1152TA	1N4148 A-TYPE
D207	30-1152TA	1N4148 A-TYPE
D208	30-1152TA	1N4148 A-TYPE
D211	30-1002	RECTI 100V1A 1N4002
D212	30-1002	RECTI 100V1A 1N4002
D213	30-1175-1TA	ZENER HZ6C3 26MM
D214	30-1147-1TA	ZENER HZ30-2 26MM
D215	30-1002	RECTI 100V1A 1N4002
D301	30-1152TA	1N4148 A-TYPE
D302	30-1152TA	1N4148 A-TYPE
D303	30-1152TA	1N4148 A-TYPE
D304	30-1152TA	1N4148 A-TYPE
D305	30-1152TA	1N4148 A-TYPE
D306	30-1152TA	1N4148 A-TYPE
D307	30-1152TA	1N4148 A-TYPE
D308	30-1152TA	1N4148 A-TYPE
D309	30-1152TA	1N4148 A-TYPE
D310	30-1152TA	1N4148 A-TYPE
D311	30-1152TA	1N4148 A-TYPE
D312	30-1136-1TA	ZENER HZ7A-2 26MM
D313	30-1152TA	1N4148 A-TYPE
D314	30-1152TA	1N4148 A-TYPE
D601	30-1152TA	1N4148 A-TYPE
D602	30-1152TA	1N4148 A-TYPE
D603	30-1152TA	1N4148 A-TYPE
D604	30-1152TA	1N4148 A-TYPE

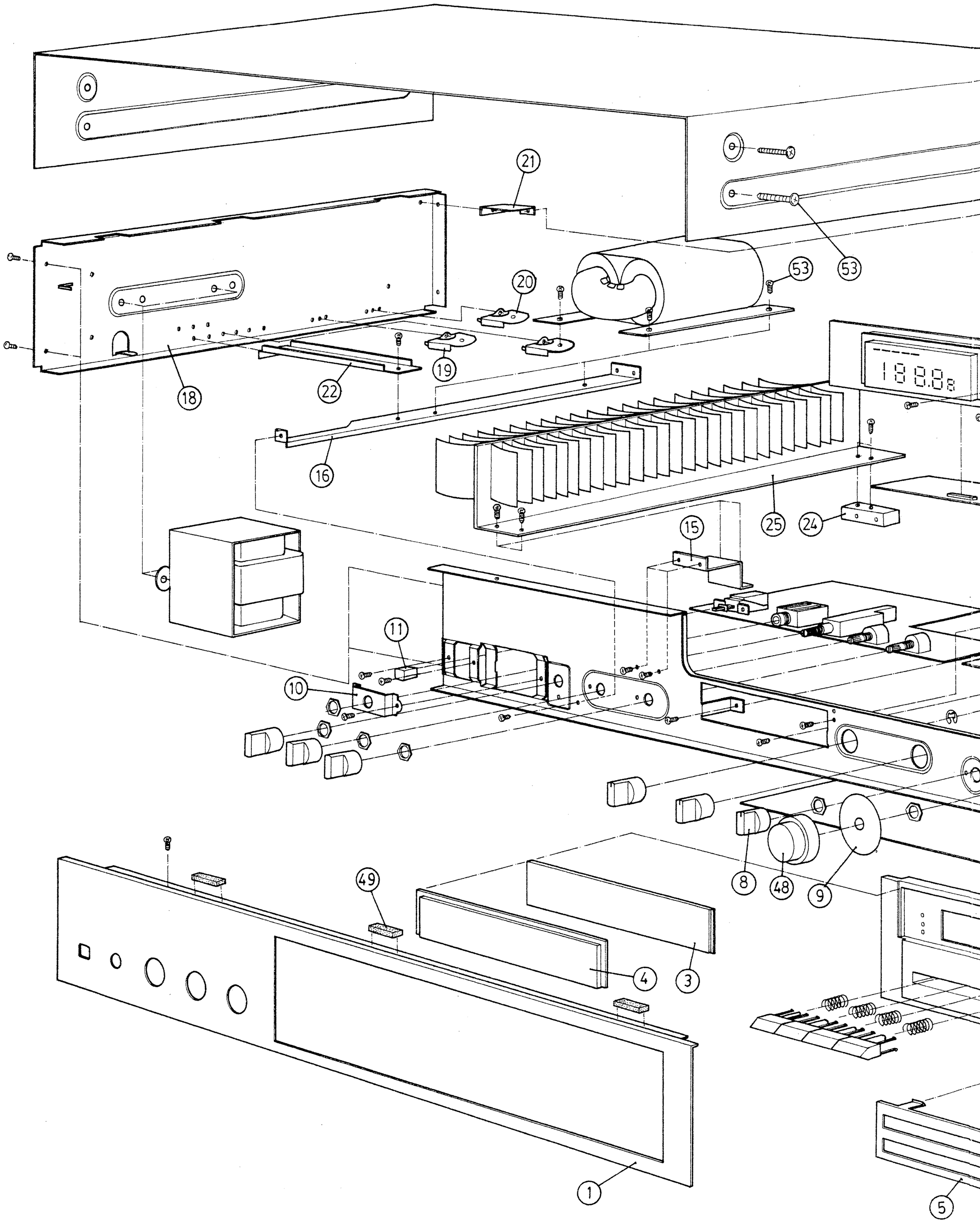
REF. NO.	PART NO.	DESCRIPTION
DIODES		
D605	30-1152TA	1N4148 A-TYPE
D606	30-1152TA	1N4148 A-TYPE
D607	30-1152TA	1N4148 A-TYPE
D608	30-1152TA	1N4148 A-TYPE
D701	30-1152TA	1N4148 A-TYPE
D702	30-1152TA	1N4148 A-TYPE
D703	30-1152TA	1N4148 A-TYPE
D704	30-1152TA	1N4148 A-TYPE
D751	30-1152TA	1N4148 A-TYPE
D752	30-1152TA	1N4148 A-TYPE
D753	30-1002A	RECTI 100V1A 1N4002/10D1
D901	30-1002A	RECTI 100V1A 1N4002/10D1
D902	30-1002A	RECTI 100V1A 1N4002/10D1
D903	30-1002A	RECTI 100V1A 1N4002/10D1
D904	30-1002A	RECTI 100V1A 1N4002/10D1
D905	30-1002A	RECTI 100V1A 1N4002/10D1
D906	30-1002A	RECTI 100V1A 1N4002/10D1
D907	30-1002A	RECTI 100V1A 1N4002/10D1
D908	30-1002A	RECTI 100V1A 1N4002/10D1
D909	30-1002A	RECTI 100V1A 1N4002/10D1
D910	30-1002A	RECTI 100V1A 1N4002/10D1
D911	30-1152TA	1N4148 A-TYPE
BD901	30-1049	BRIDGE 200V3A KBL02
BD902	30-1049	BRIDGE 200V3A KBL02
ZENER DIODES		
ZD601	30-1041-1TA	HZ22-2 26MM
ZD602	30-1041-1TA	HZ22-2 26MM
ZD701	30-1044-1TA	HZ15-2 26MM
ZD702	30-1044-1TA	HZ15-2 26MM
ZD901	30-1041-1TA	HZ22-2 26MM
ZD902	30-1041-1TA	HZ22-2 26MM
ZD903	30-1044-1TA	HZ15-2 26MM
ZD904	30-1044-1TA	HZ15-2 26MM
ZD905	30-1044-1TA	HZ15-2 26MM
ZD906	30-1044-1TA	HZ15-2 26MM
INTEGRATED CIRCUITS		
IC101	30-3035	IC FM TA7060P&AP
IC102	30-3206	IC FM SYSTEM LA1235
IC103	30-3008	IC D-TYPE FLOP CD4013
IC104	30-3309	IC METER DRIVER CI6125E
IC105	30-3346	IC MPX LA3410
IC106	30-3069	IC DUAL OP RC4558
IC201	30-3149	IC FM PRESCALLER TD6104P
IC202	30-3342	IC SYNTHESIS TC9157AP
IC203	30-3310	IC DISPLAY DRIVER TC9159P
IC401	30-3195	IC AM SYSTEM LA1245
IC501	30-3247	IC DUAL OP M5219L
IC751	30-3187	IC PROTECTION TA7317P
IC801	30-3247	IC DUAL OP M5219L
IC802	30-3247	IC DUAL OP M5219L
TRANSISTORS		
Q001	30-2379	FET MOS 3SK114-0
Q002	30-2298	FET 2SK55
Q003	30-2298	FET 2SK55
Q004	30-2298	FET 2SK55
Q005	30-2298	FET 2SK55
Q101	30-2298	FET 2SK55
Q102	30-2019	TR 2SC930
Q201	30-2161	FET 2SK40

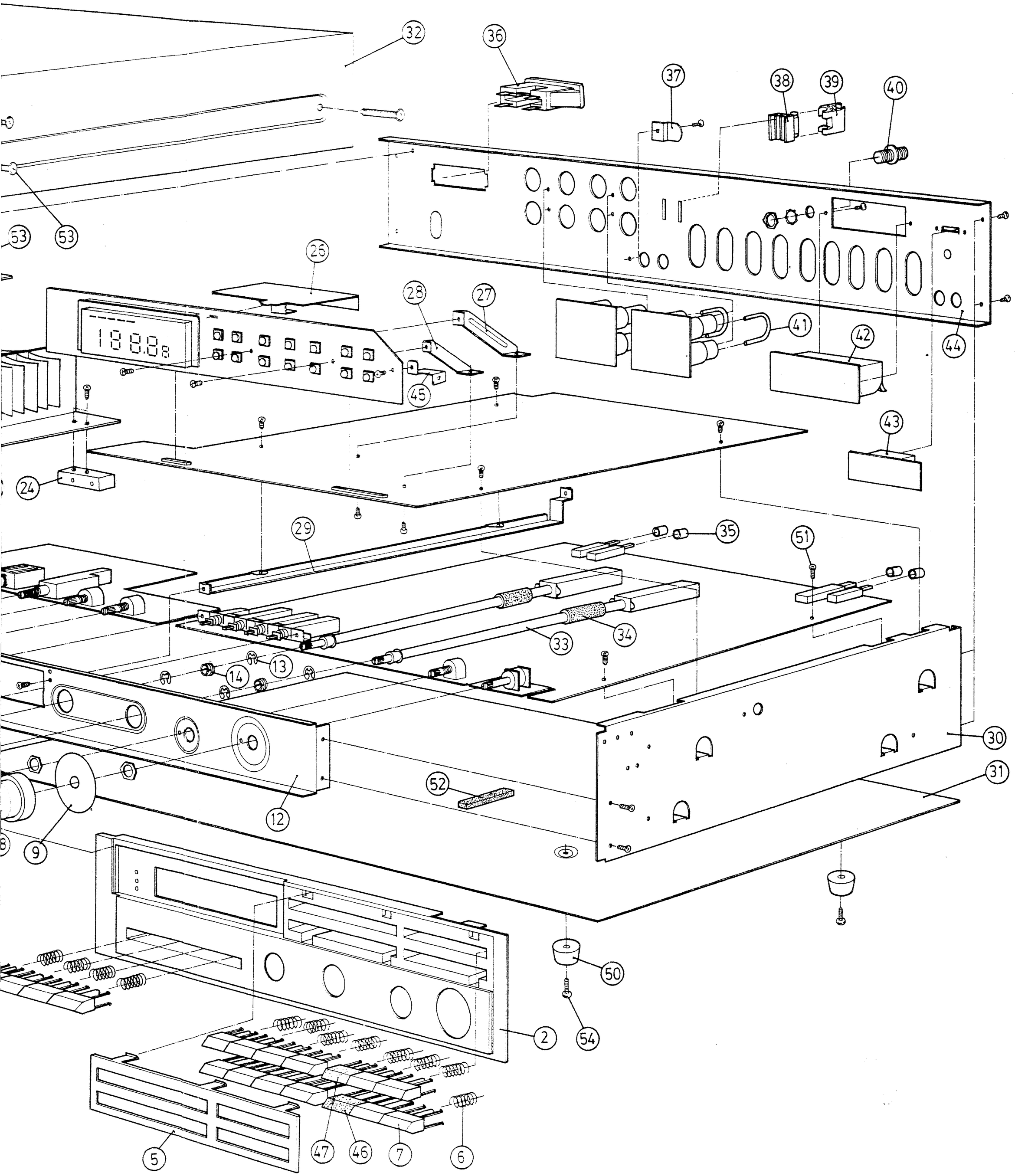
REF. NO.	PART NO.	DESCRIPTION
TRANSISTORS		
Q202	30-2156	TR 2SC1815GR
Q203	30-2156	TR 2SC1815GR
Q204	30-2171	TR 2SA733P
Q205	30-2156	TR 2SC1815GR
Q206	30-2171	TR 2SA733P
Q207	30-2156	TR 2SC1815GR
Q208	30-2156	TR 2SC1815GR
Q209	30-2156	TR 2SC1815GR
Q210	30-2353	TR BC639
Q211	30-2353	TR BC639
Q212	30-2353	TR BC639
Q213	30-2353	TR BC639
Q214	30-2353	TR BC639
Q215	30-2078	TR 2SD330E
Q216	30-2156	TR 2SC1815GR
Q219	30-2156	TR 2SC1815GR
Q220	30-2156	TR 2SC1815GR
Q221	30-2156	TR 2SC1815GR
Q222	30-2156	TR 2SC1815GR
Q223	30-2156	TR 2SC1815GR
Q301	30-2171	TR 2SA733P
Q302	30-2171	TR 2SA733P
Q303	30-2156	TR 2SC1815GR
Q304	30-2156	TR 2SC1815GR
Q305	30-2156	TR 2SC1815GR
Q306	30-2156	TR 2SC1815GR
Q307	30-2156	TR 2SC1815GR
Q308	30-2156	TR 2SC1815GR
Q310	30-2156	TR 2SC1815GR
Q311	30-2171	TR 2SA733P
Q312	30-2359	TR 2SD655D
Q313	30-2359	TR 2SD655D
Q401	30-2019	TR 2SC930
Q501	30-2349	TR 2SC2602G/H
Q502	30-2349	TR 2SC2602G/H
Q503	30-2375	TR 2SA1114G/H
Q504	30-2375	TR 2SA1114G/H
Q505	30-2349	TR 2SC2602G/H
Q506	30-2349	TR 2SC2602G/H
Q601	30-2090-2	TR BC546B
Q602	30-2090-2	TR BC546B
Q603	30-2090-2	TR BC546B
Q604	30-2090-2	TR BC546B
Q605	30-2259	TR 2SB649C/2SB649D
Q606	30-2259	TR 2SB649C/2SB649D
Q607	30-2259	TR 2SB649C/2SB649D
Q608	30-2259	TR 2SB649C/2SB649D
Q609	30-2260	TR 2SD669C/2SD669D
Q610	30-2260	TR 2SD669C/2SD669D
Q611	30-2260	TR 2SD669C/2SD669D
Q612	30-2260	TR 2SD669C/2SD669D
Q613	30-2260	TR 2SD669C/2SD669D
Q614	30-2260	TR 2SD669C/2SD669D
Q615	30-2259	TR 2SB649C/2SB649D
Q616	30-2259	TR 2SB649C/2SB649D
Q617	30-2399	TR 2SC3519
Q618	30-2399	TR 2SC3519
Q619	30-2398	TR 2SA1386
Q620	30-2398	TR 2SA1386
Q651	30-2337-1	TR BC337-25
Q652	30-2342	TR BC327-25
Q653	30-2260	TR 2SD669C/2SD669D

REF. NO.	PART NO.	DESCRIPTION
TRANSISTORS		
Q654	30-2259	TR 2SB649C/2SB649D
Q655	30-2384	TR 2SC2837
Q656	30-2383	TR 2SA1186
Q657	30-2384	TR 2SC2837
Q658	30-2383	TR 2SA1186
Q659	30-2096	TR BC556B
Q751	30-2237	TR BF422
Q752	30-2237	TR BF422
Q753	30-2096	TR BC556B
Q801	30-2090-2	TR BC546B
Q802	30-2090-2	TR BC546B
Q803	30-2096	TR BC556B
Q804	30-2096	TR BC556B
Q805	30-2090-2	TR BC546B
Q806	30-2090-2	TR BC546B
Q807	30-2096	TR BC556B
Q808	30-2096	TR BC556B
Q901	30-2260	TR 2SD669C/2SD669D
Q902	30-2259	TR 2SB649C/2SB649D
Q903	30-2090-2	TR BC546B
Q904	30-2096	TR BC556B
Q905	30-2353	TR BC639
Q906	30-2354	TR BC640
Q907	30-2090-2	TR BC546B
Q908	30-2096	TR BC556B
Q909	30-2260	TR 2SD669C/2SD669D
COILS & IFT		
L001	29-1099*A	COIL VARI 22pF +-6% 100MHz
L002	29-1241	COIL VARI 39pF +-6% 100MHz
L003	29-1241	COIL VARI 39pF +-6% 100MHz
L004	29-1051	COIL 2.2uH +-10% 7.96MHz
L005	29-1147	COIL VARI 36pF +-6% 100MHz
L006	29-1104*B	COIL 15uH +-10% 2.52MHz
L007	29-1051	COIL 2.2uH +-10% 7.56MHz
L101	29-1102*B	COIL 15uH +-10% 2.52MHz
L103	29-1102*B	COIL 15uH +-10% 2.52MHz
L104	29-1092*B	COIL 12uH +-10% 2.52MHz
IFT001	29-3111	IFT MIXER 119AC-AP17BV/41K7H5
IFT101	29-3190	IFT FM IF DET L16-400-1182
IFT102	29-3221	ANTI BIRDY FILTER KM-10DB 5305
LD201	29-3049	LPF FM MPX LA1196
LD2012	29-3049	LPF FM MPX LA1196L
T401	29-3109	IFT AM OSC 7BR-BD119Y
T402	29-3029	IFT AM WE 10AI/21K10H/10EZ
T403	29-3137	CFT AM CFTS-167A
T404	29-3189	COIL AM RT
L401	29-1246	COIL 400mH +-5%
TRIMER & CERAMIC FILTER		
TC001	17-1040	CAP TRIM CER ECR-HA010A11A
TC002	17-1040	CAP TRIM CER ECR-HA010A11A
TC003	17-1040	CAP TRIM CER ECR-HA010A11A
TC004	17-1040	CAP TRIM CER ECR-HA010A11A
TC005	17-1040	CAP TRIM CER ECR-HA010A11A
TC006	17-1040	CAP TRIM CER ECR-HA010A11A
CF101	29-3128	CF SFE10.7MLH-Z
CF102	29-3128	CF SFE10.7MLH-Z
CF103	29-3128	CF SFE10.7MLH-Z
CF401	29-3123	CER FILTER BFU450KX2
CR101	29-3212	CEAMIC RESONATOR CSB456F11

REF. NO.	PART NO.	DESCRIPTION
MISCELIANEOUS		
CY201	30-4004	XTAL 7.2MHz
	31-3011	DISPLAY LED
	29-2132	X'FMR POWER
	29-2140	X'FMR POWER
	CN-11-LD2509	9P PLUG P=2.5MM STRAIGHT
	CN-11-LP2518	18P PLUG P=2.5MM STRAIGHT
	CN-07-SP2502	2P PLUG P=2.5MM STRAIGHT
	CN-07-SP2503	3P PLUG P=2.5MM STRAIGHT
CM2	11-5077	HEAT SINK
COMMON	18-1030	BALUN CORE
90-1360	W9-0E150C10-10	COAXIAL 1.5C-2V 75 OHM 150MM
90-1360	W9-0E200C10-10	COAXIAL 1.5C-2V 75 OHM 200MM
90-1360	11-9120	SHIELD FE P930 SPTE
90-1360	11-9123*A	SHIELD FE SPACER A
90-1360	11-9124	SPACER FE B P930 SPTE
90-1360	11-9142	FE COVER B
90-1360	11-4061	SHIELD PLATE FOR IC TD6104P
SW1	31-1275	SW PUSH SPJ242NE01-FL
SW2	31-1276	SW PUSH SPJ222NE01-FL
SW3	31-1079F	SW SLIDE SSB23FL025
SW4	31-1345	SW ROT SRS026N02-FL
SW5	31-1346	SW ROT SRS046N08-FL
SW6	31-1343	SW PUSH ESB-62
SW7	31-1343	SW PUSH ESB-62
SW8	31-1343	SW PUSH ESB-62
SW9	31-1276	SW PUSH SPJ222NE01
SW10	31-1276	SW PUSH SPJ222NE01
SW11	31-1344	SW ROT SRZV044N12-FL
RELAY 1	35-3051	RELAY OMI-SS-2180
RESISTORS (All are carbon film 1/4W +-5% unless noted othise)		
R001	16-1/4CA473J	RES CA 47K OHM 1/4W
R002	16-1/4CA473J	RES CA 47K OHM 1/4W
R003	16-1/4CA470J	RES CA 47 OHM 1/4W
R004	16-1/4CA473J	RES CA 47K OHM 1/4W
R005	16-1/4CA101J	RES CA 100 OHM 1/4W
R006	16-1/4CA330J	RES CA 33 OHM 1/4W
R007	16-1/4CA470J	RES CA 47 OHM 1/4W
R008	16-1/4CA473J	RES CA 47K OHM 1/4W
R009	16-1/4CA102J	RES CA 1K OHM 1/4W
R010	16-1/4CA103J	RES CA 10K OHM 1/4W
R011	16-1/4CA101J	RES CA 100 OHM 1/4W
R012	16-1/4CA103J	RES CA 10K OHM 1/4W
R013	16-1/4CA101J	RES CA 100 OHM 1/4W
R014	16-1/4CA101J	RES CA 100 OHM 1/4W
R016	16-1/4CA104J	RES CA 100K OHM 1/4W
R017	16-1/4CA273J	RES CA 27K OHM 1/4W
R018	16-1/4CA330J	RES CA 33 OHM 1/4W
R019	16-1/4CA104J	RES CA 100K OHM 1/4W
R020	16-1/4CA105J	RES CA 1M OHM 1/4W

ASSEMBLY DIAGRAM

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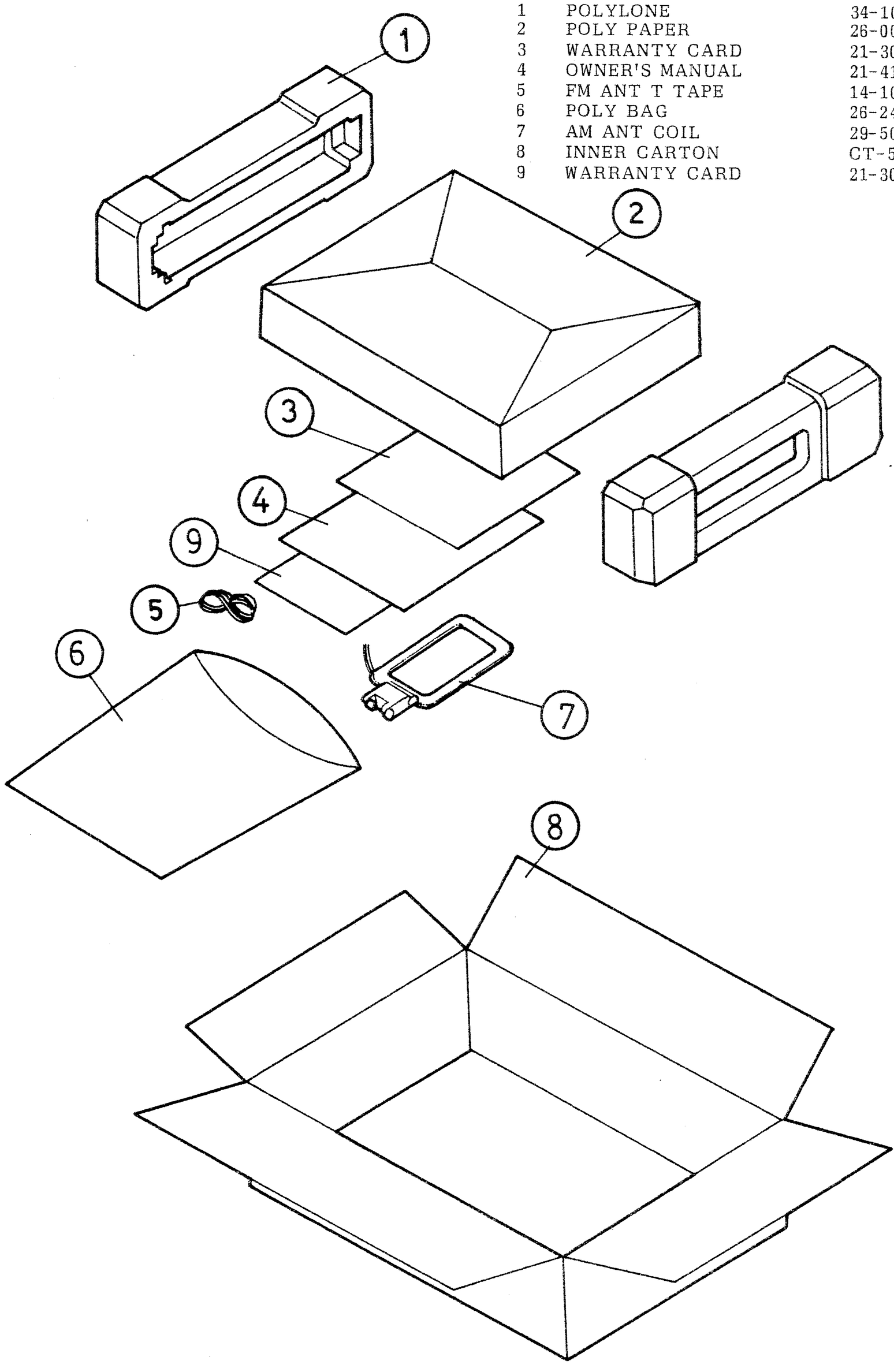
MECHANICAL PARTS LIST

ITEM	DESCRIPTION	PARTS NO.	Q'TY
1	FRONT PANEL	11-8365	1
2	AUX PANEL	11-8366	1
3	L.E.D. WINDOW	11-8364	1
4	ACPYLIC WINDOW	11-8363	1
5	FRONT COVER	11-1179	1
6	SPRING B	12-5059	18
7	PUSH KNOB	12-3180	16
8	KNOB FOR BASS TREBLE	12-3T10*A	6
9	MASKER	28-2120	1
10	HEADPHONE'S BRACKET	11-2352	1
11	PUSH SW KNOB	12-3T07	1
12	FRONT CHASSIS	11-6197	1
13	E-RING	15-5014	4
14	TUBE FOR SHAFT	13-4027*A	2
15	BRACKET B	11-2365	1
16	CENTER CHASSIS A	11-6194	1
18	LEFT CHASSIS	11-6193	1
19	BRACKET FOR X'FMR B	11-2402	1
20	BRACKET FOR X'FMR	11-2382	2
21	BRACKET	11-2335	1
22	X'FMR PLATE	11-8379	1
24	HEAT SINK BRACKET	11-2397	1
25	HEAT SINK	11-5079	1
26	BRACKET G	11-2401	1
27	BRACKET D	11-2383	1
28	BRACKET FOR PCB	11-2396	1
29	CENTER CHASSIS B	11-6195	1
30	RIGHT CHASSIS	11-6196	1
31	BOTTOM CHASSIS	11-6192	1
32	CABINET	50-1054	1
33	SHAFT	11-7081	2
34	TUBE	12-7003*A	2
35	POWER SW KNOB	12-3T02	4
36	RECEPTACLE	12-2193	1
37	BRACKET F	11-2395	1
38	RETAINER A FOR AM LOOP	28-2113	1
39	RETAINER B FOR AM LOOP	28-2114	1
40	CON NUT	12-1032	1
41	JUMPER FOR IN/ONT	11-4011	2
42	4P PUSH TERMINAL	12-2177	1
43	SW SLIDE	31-1079F	1
44	BACK PANEL	11-8367	1
45	BRACKET E	11-2384	1
46	PUSH KNOB	11-3180-2	1
47	PUSH KNOB	11-3180-1	1
48	VOLUME KNOB	12-3168*A	1
49	RETAINER	28-2090	6
50	RUBBER FEET	28-1029	4
51	SPECIAL SCREW	15-2075	8
52	SPONGE	28-2135	1
53	SPECIAL SCREW	15-2062	14
54	STANDER'S SCREW	15-2063	4

PACKING DIAGRAM

ITEM	DESCRIPTION	PART NO.	Q'TY
1	POLYLONE	34-1072	2
2	POLY PAPER	26-0005*B	1
3	WARRANTY CARD	21-3037*A	1
4	OWNER'S MANUAL	21-4132	1
5	FM ANT T TAPE	14-1027	1
6	POLY BAG	26-2434*A	1
7	AM ANT COIL	29-5023	1
8	INNER CARTON	CT-5272	1
9	WARRANTY CARD	21-3051	1



PROTON TECHNICAL BULLETIN

MODEL NUMBER: D940 NO: 88-6

SUBJECT: AUDIBLE HUM IN OUTPUTS DATE: June 28, 1988

APPLICABLE SERIAL NUMBERS:

It has been noted that certain D940's exhibit 25-50mv hum at approximately 120hz out of one or both channels with no source selected, volume control at minimum and treble control at detent position. To correct this condition, we can offer the following modification:

1) Remove Top and Bottom covers.

2) From top of unit, front panel facing you, remove screw from side chassis - holding wire to tuner shield case.

3) Remove (3) screws holding top board (ref# 19-1360) in place.

4) Remove 'black' screw on right side chassis holding (19-1360) in place. (The one with locktight)

5) Disconnect any quick release connects which will restrict removal of board (19-1360).

6) Gently lift & pivot top board out of the way (19-1360). Now you have access to bottom board.

Now you will notice that two (2) flat ribbon cables (6 Pin each) originating from the Treble Pot and right below it are routed under the rotating switching shafts for the Record and Listen selector switches and terminating at W.A.E. and W.A.F. on bottom board (ref# 19-1361). Proceed as follows:

A) Desolder the (2) flat ribbon cables from their respective points (W.A.E. & W.A.F.) and route them out of the way. Also remove silicone glue.

B) Desolder and remove C943, C944, C945 and C946. (Currently .01uf 50V +-5%)

C) Install .1uf 50V +- 5% (new value) for C943, C944, C945 and C946.

D) Now that new Caps are installed, reroute the (2) flat ribbon cables over the Rec/Listen rotating shafts - taking care to route them under the bracket which holds (ref# 19-1360 - top bd) in place. This step is absolutely crucial for the mod to work! Resolder ribbon cables to W.A.E. and W.A.F., add silicone glue.

E) Reinstall top board (19-1360) in reverse order which you removed.

F) Replace Top and Bottom covers.

Now recheck noise at speaker and headphone out. (Typically 5mv)

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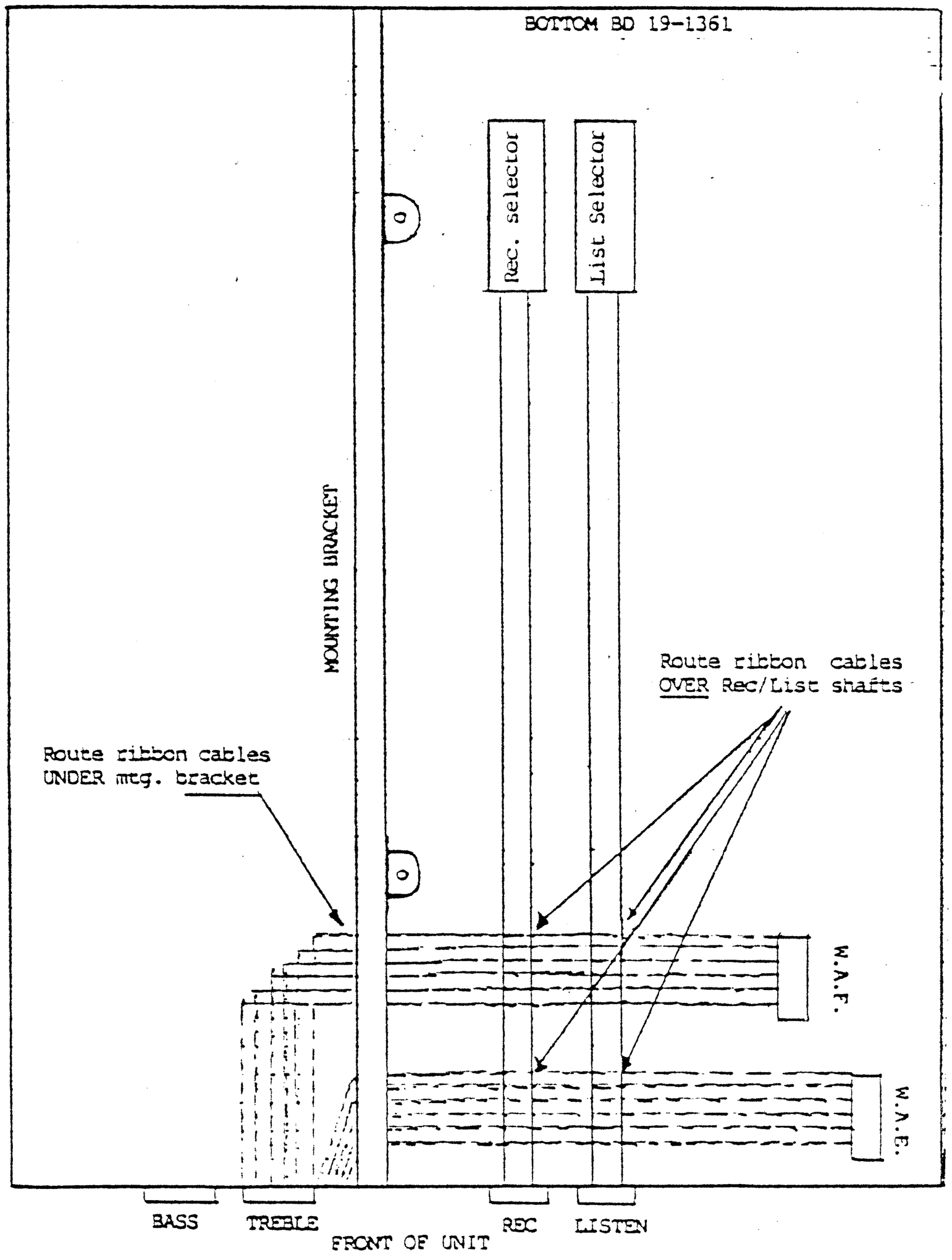
PROTON TECHNICAL BULLETIN

MODEL NUMBER: D940 NO: 88-6

SUBJECT: AUDIBLE HUM IN OUTPUTS DATE: JUNE 28, 1988

APPLICABLE SERIAL NUMBERS: PAGE 2 OF 2



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PROTON

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